

## PURCHASE DESCRIPTION

UNIFORM, ENHANCED FLAME RESISTANT COMBAT ENSEMBLE (EFRCE)  
BLOUSE AND TROUSER

## 1. SCOPE

1.1 Scope. This document covers the requirements for the enhanced flame-resistant combat ensemble (EFRCE), blouse and trouser. The uniform is used to provide flame resistance and insect protection needed for wear in field and combat operations.

1.2 Classification. The EFRCE blouse and trousers shall be of one type in the following classes, as specified (see 6.2).

1.2.1 Type.

Type I - EFRCE Blouse and Trouser with Durable Insect Protection

1.2.2 Classes.

Class 1 Woodland MARPAT Camouflage Printed

Class 2 Desert MARPAT Camouflage Printed

Class 3 NWU II Desert Digital Camouflage Printed

Class 4 NWU III Woodland Digital Camouflage Printed

1.2.2.1 Schedule of Sizes. The blouse and trousers shall be constructed in the following sizes, as specified (see 6.2).

## SCHEDULE OF SIZES – BLOUSE

| X-Small | Small    | Medium   | Large   | X-Large | XX-Large |
|---------|----------|----------|---------|---------|----------|
|         | XX-Short | XX-Short |         |         |          |
| X-Short | X-Short  | X-Short  | X-Short |         |          |
| Short   | Short    | Short    | Short   | Short   |          |
| Regular | Regular  | Regular  | Regular | Regular | Regular  |
|         | Long     | Long     | Long    | Long    | Long     |
|         | X-Long   | X-Long   | X-Long  | X-Long  | X-Long   |
|         |          | XX-Long  | XX-Long | XX-Long | XX-Long  |

Beneficial comments, recommendations, additions, deletions, clarifications, and any other data that may improve this document should be sent to: Marine Corps Systems Command [USMC\\_STDZ@usmc.mil](mailto:USMC_STDZ@usmc.mil), with the subject line "Document Comment".

## SCHEDULE OF SIZES – TROUSERS

| X-Small | Small   | Medium  | Large   | X-Large | XX-Large |
|---------|---------|---------|---------|---------|----------|
| X-Short | X-Short | X-Short |         |         |          |
| Short   | Short   | Short   | Short   |         |          |
| Regular | Regular | Regular | Regular | Regular | Regular  |
| Long    | Long    | Long    | Long    | Long    | Long     |
|         | X-Long  | X-Long  | X-Long  | X-Long  | X-Long   |
|         |         | XX-Long | XX-Long |         | XX-Long  |

## 2. APPLICABLE PUBLICATIONS.

2.1 General. The publications listed in this section are specified in sections 3, 4 or 5 of this specification. This section does not include publications cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of publications cited in sections 3, 4 and 5 of this specification, whether or not they are listed.

2.2 Government publications.

2.2.1 Government specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these publications are those cited in the solicitation or contract.

## COMMERCIAL ITEM DESCRIPTIONS

|           |   |                                                 |
|-----------|---|-------------------------------------------------|
| A-A-50198 | - | Thread, Gimp, Cotton, Buttonhole                |
| A-A-55126 | - | Fastener Tapes, Hook and Loop, Synthetic        |
| A-A-55195 | - | Thread: Para-Aramid, Spun, Intermediate Modulus |
| A-A-55217 | - | Thread, Aramid, Spun Staple                     |
| A-A-55634 | - | Zippers (Fasteners, Slide Interlocking)         |

## DEPARTMENT OF DEFENSE SPECIFICATIONS

|               |   |                                                     |
|---------------|---|-----------------------------------------------------|
| MIL-DTL-44411 | - | Insect Repellent, Permethrin                        |
| MIL-PRF-26514 | - | Polyurethane Foam, Rigid or Flexible, for Packaging |
| MIL-C-44296   | - | Cloth, Fusibles                                     |
| MIL-W-5664D   | - | Webbing, Textile, Elastic                           |

(Copies of these publications are available online at [www.quicksearch.dla.mil](http://www.quicksearch.dla.mil).)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those specified in the solicitation or contract (see 6.2).

| Drawing Number |   | Drawing Description                                       |
|----------------|---|-----------------------------------------------------------|
| 2-1-2525       | - | Woodland MARPAT- 4 color (Coyote 476)                     |
| 2-1-2526       | - | Woodland MARPAT- 4 color (Green 474 with EGA symbol)      |
| 2-1-2527       | - | Woodland MARPAT- 4 color (Black 477)                      |
| 2-1-2528       | - | Woodland MARPAT- 4 color (Khaki 475)                      |
| 2-1-2529       | - | Desert MARPAT- 4 color (Light Tan 479)                    |
| 2-1-2530       | - | Desert MARPAT- 4 color (Urban Tan 478)                    |
| 2-1-2531       | - | Desert MARPAT- 4 color (Light Coyote 481 with EGA symbol) |

- 2-1-2532 - Desert MARPAT- 4 color (Highland 480)
- 2-1-2533 - Woodland MARPAT- 3 color (Coyote 476)
- 2-1-2534 - Woodland MARPAT- 3 color (Green 474)
- 2-1-2535 - Woodland MARPAT- 3 color (Khaki 476)

(Copies of drawings are available from the U.S. Army Developmental Command – Soldier Center, Natick Soldier Center, Natick, MA 01760)

Commercial US Governmentally Controlled Performance Specification Camouflage Print  
Performance Specification for AOR 1, AOR 2, NWU II, and NWU III.

(Copies are available from Program Manager – Special Operations Forces (SOF) Survival, Support and Equipment Systems, Natick Soldier Center, Natick, MA 01760.)

#### DEFENSE LOGISTICS AGENCY

DLA Troop Support Clothing & Textiles Additional Quality Assurance Requirements

(Copies of this publication may be obtained from [www.dla.mil](http://www.dla.mil).)

#### CODE OF FEDERAL REGULATIONS

- Title 40, part 798.4500 - Primary Eye Irritation
- Title 40, part 798.4100 - Dermal Sensitization
- Title 40, part 798.4470 - Primary Dermal Irritation

(Copies are available online at [www.ecfr.gov](http://www.ecfr.gov))

#### ENVIRONMENTAL PROTECTION AGENCY (EPA)

EPA Product Performance Test Guidelines OPPTS 810.370 Insect Repellents for Human Skin and Outdoor Premises

FIFRA as amended by the Food Quality Protection Act of 1996 and the Pesticide Registration Improvement Act of 2003.

(Copies of these publications are available online at <http://www.epa.gov> )

#### FEDERAL TRADE COMMISSION

- 15 U.S.C. § 70 - TEXTILE FIBER PRODUCTS IDENTIFICATION ACT

(Copies of these publications are available online at <http://www.ftc.gov>.)

#### US DEPARTMENT OF AGRICULTURE (USDA)

Bite Protection Test Methodology, Enhanced Flame Resistant Combat Ensemble

(Copies of this publication can be requested through USDA Southeast Area, P.O. Box 225, Stoneville MS, 38776)

2.3 Non-Government publications. The following publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these publications are those specified in the solicitation or contract (see 6.2).

#### AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS

- AATCC TM8 - Colorfastness to Crocking: Crockmeter Method
- AATCC TM15 - Colorfastness to Perspiration
- AATCC TM16.2 - Colorfastness to Light: Carbon-Arc
- AATCC TM16.3 - Colorfastness to Light: Xenon-Arc

- AATCC TM20A - Fiber Analysis: Quantitative
- AATCC TM61 - Colorfastness to Laundering, Home and Commercial: Accelerated
- AATCC TM81 - pH of the Water Extracted from Wet Processed Textiles
- AATCC TM96 - Dimensional Changes in Commercial Laundering of Woven and Knitted Fabrics Except Wool
- AATCC TM100 - Antibacterial Finishes on Textile Materials
- AATCC TM135 - Assessment of Dimensional Changes of Fabrics after Home Laundering
- AATCC EP6 - Instrumental Color Measurement
- AATCC EP9 - Visual Assessment of Color Difference of Textiles

(Copies of these publications are available online at [www.aatcc.org](http://www.aatcc.org).)

#### AMERICAN SOCIETY FOR TESTING AND MATERIALS

- ASTM D76 - Standard Specification for Tensile Testing Machines for Textiles
- ASTM E96 - Standard Test Methods for Water Vapor Transmission of Materials
- ASTM D737 - Standard Test Method for Air Permeability of Textile Fabrics
- ASTM D1424 - Standard Test Method for Tearing Strength of Fabrics by Falling Pendulum Type (Elmendorf) Apparatus
- ASTM F1930 - Standard Test Method for Evaluation of Flame Resistant Clothing for Protection Against Fire Simulations Using an Instrumented Manikin
- ASTM D2256 - Standard Test Method for Tensile Properties of Yarns by the Single-Strand Method
- ASTM D3511 - Standard Test Method for Pilling Resistance and Other Related Surface Changes of Textile Fabrics: Brush Pilling Tester
- ASTM D3776/D3776M - Standard Test Method for Mass Per Unit Area (Weight) of Fabric
- ASTM D5034 - Standard Test Method for Breaking Strength and Elongation of Textile Fabrics (Grab Test)
- ASTM D6193 - Standard Practice for Stitches and Seams
- ASTM D6413 - Standard Test Method for Flame Resistance of Textiles (Vertical Test)
- ASTM D6797 - Fabrics, Bursting Strength of, Constant-Rate-of-Extension (CRE) Ball Burst Test

(Copies may be obtained online at [www.astm.org](http://www.astm.org).)

#### NATIONAL FIRE PROTECTION ASSOCIATION

- NFPA 1971 - Standard on Protective Ensembles for Structural Fire Fighting and Proximity Fire Fighting

(Copies may be obtained online at [www.nfpa.org](http://www.nfpa.org).)

#### AMERICAN SOCIETY FOR QUALITY

- ASQ/ANSI Z1.4 - Sampling Procedures and Tables for Inspection by Attribute

(Copies may be obtained online at [www.asq.org](http://www.asq.org).)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

### 3. REQUIREMENTS

#### 3.1 Inspections.

3.1.1 First article. When specified, (see 6.2), a sample shall be subjected to first article inspection (see 6.3), in accordance with 4.2.

3.1.2 Quality conformance inspection. When specified, (see 6.2), a sample shall be subjected to quality conformance inspection, in accordance with 4.3.

3.2 Recycled, recovered, environmentally preferable, or biobased materials. Unless otherwise specified in the contract or order (see 6.2), recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Material. When produced for the government, each roll of finished cloth shall be labeled or ticketed for fiber content in accordance with the Rules and Regulations under the Textile Fiber Products Identification Act. Each roll shall indicate cloth class.

3.3.1 Basic material, woven. The base material for construction of the blouse sleeves, yoke and collar; and all parts of the trouser, shall consist of a flame-resistant cotton/nylon/meta-aramid twill cloth conforming to the requirements as specified table I when tested as specified in 4.4.5. Fabric shall be printed with the appropriate camouflage pattern (as specified in the solicitation).

TABLE I. Material Requirements for Cloth, Woven.

| CHARACTERISTIC                                                     | REQUIREMENT |
|--------------------------------------------------------------------|-------------|
| Weight, ounces per square yard                                     | 6.4 – 7.1   |
| Fiber content, percent ( $\pm 3\%$ )                               |             |
| Meta-aramid                                                        | 42          |
| Nylon                                                              | 31          |
| Cotton                                                             | 27          |
| Weave                                                              | Twill       |
| Breaking Strength, pounds (minimum),<br>after one laundering cycle |             |
| Dry                                                                |             |
| Warp                                                               | 165         |
| Fill                                                               | 150         |
| Wet                                                                |             |
| Warp                                                               | 140         |
| Fill                                                               | 125         |

TABLE I. Material Requirements for Cloth, Woven. Continued

| CHARACTERISTIC                                                                             | REQUIREMENT  |
|--------------------------------------------------------------------------------------------|--------------|
| Breaking Strength, pounds (minimum), after 25 laundering cycles                            |              |
| Dry                                                                                        |              |
| Warp                                                                                       | 150          |
| Fill                                                                                       | 150          |
| Wet                                                                                        |              |
| Warp                                                                                       | 140          |
| Fill                                                                                       | 125          |
| Air Permeability, cubic feet/minute (minimum)                                              | 20           |
| Moisture Vapor Transmission Rate, grams per square meter per 24 hours (minimum)            | 1000         |
| Pilling Appearance after 5 laundering cycles, rating (minimum)                             | 3            |
| Tear Strength, pounds (minimum)<br>after 1 laundering cycle and after 25 laundering cycles |              |
| Dry                                                                                        |              |
| Warp                                                                                       | 12           |
| Fill                                                                                       | 16           |
| Wet                                                                                        |              |
| Warp                                                                                       | 12           |
| Fill                                                                                       | 16           |
| Colorfastness (minutes)                                                                    |              |
| Laundering (after 4 cycles)                                                                | 4-5          |
| Light (40 AFU)                                                                             | 4            |
| Perspiration (acid and alkaline)                                                           |              |
| Color change                                                                               | 4-5          |
| Staining                                                                                   | 4-5          |
| Crocking                                                                                   |              |
| Dry                                                                                        | 4 <u>1</u> / |
| Wet                                                                                        | 3 <u>1</u> / |
| Dimensional Stability, percent (maximum) after commercial laundering, 5 cycles             |              |
| Warp                                                                                       | 5.0          |
| Fill                                                                                       | 5.0          |
| Drying Time, minutes (maximum)                                                             | 90           |
| Thermal Protective Performance, calories per square centimeter (minimum) with spacer       | 10           |
| Thermal Shrinkage, percent (maximum)                                                       |              |
| Warp and fill                                                                              | 10           |

TABLE I. Material Requirements for Cloth, Woven. Continued

| CHARACTERISTIC                             | REQUIREMENT |
|--------------------------------------------|-------------|
| Vertical Flame, Initial                    |             |
| After Flame, seconds (maximum)             | 2           |
| Char Length, inches (maximum)              | 5           |
| Melt/Drip                                  | None        |
| Vertical Flame, after 25 laundering cycles |             |
| After Flame, seconds (maximum)             | 2           |
| Char Length, inches (maximum)              | 5           |
| Melt/Drip                                  | None        |

1/ Finished cloth fastness to crocking for Black shall have a rating not lower than 1.5.

3.3.2 Basic material, knit. The basic material for the blouse torso for all classes of shirt shall be a meta-aramid/flame resistant rayon/nylon/para-aramid knit fabric (see 6.4). Unless otherwise specified in the contract or order (see 6.2), the knit shall be a jersey knit and dyed to coyote 498, and conform to the requirements as specified in table II.

TABLE II. Material Requirements for Cloth, Knit.

| CHARACTERISTIC                                         | REQUIREMENT |
|--------------------------------------------------------|-------------|
| Weight, ounces per square yard                         | 5.3 – 6.3   |
| Fiber content, percent ( $\pm 3\%$ )                   |             |
| Meta-aramid                                            | 45          |
| FR Rayon                                               | 32          |
| Nylon                                                  | 17          |
| Para-aramid                                            | 6           |
| Knit construction                                      | See 3.3.2   |
| Bursting Strength, pounds (min)                        |             |
| Initial (no laundering) and after 25 laundering cycles |             |
| Dry                                                    | 100         |
| Wet                                                    | 75          |
| Air Permeability, cfm (min)                            | 200         |
| Microbial growth, percent reduction                    |             |
| Initial (no laundering) and after 25 laundering cycles |             |
| <i>Staphylococcus aureus</i>                           |             |
| <i>Pseudomonas aeruginosa</i>                          | 90          |
| Dimensional Stability, Commercial                      |             |
| Laundering 5 cycles percent, (maximum)                 |             |
| Warp                                                   | 8.0         |
| Fill                                                   | 8.0         |
| Drying Time, minutes (max)                             | 90          |

TABLE II. Material Requirements for Cloth, Knit. Continued

| CHARACTERISTIC                                                            | REQUIREMENT |
|---------------------------------------------------------------------------|-------------|
| Colorfastness                                                             |             |
| Laundering (after 4 cycles)                                               | 4           |
| Light (40 hours)                                                          | 4           |
| Perspiration                                                              |             |
| Color change                                                              | 4-5         |
| Staining                                                                  | 4-5         |
| Crocking                                                                  |             |
| Wet                                                                       | 4           |
| Dry                                                                       | 4           |
| Moisture Vapor Transmission, per square meter per 24 hours (min)          | 1800        |
| Pilling Appearance after 5 laundering cycles (min)                        | 3           |
| Thermal Protective Performance, cal/cm <sup>2</sup> (minimum) with spacer | 8           |
| Vertical Flame, Initial                                                   |             |
| After Flame, seconds (maximum)                                            | 2           |
| Char Length, inches (maximum)                                             | 5           |
| Melt/Drip                                                                 | None        |
| Vertical Flame, After 25 Laundering Cycles                                |             |
| After Flame, seconds (max)                                                | 2           |
| Char Length, inches (max)                                                 | 5           |
| Melt/Drip                                                                 | None        |

3.3.3 Toxicity. All finished cloth and garments shall not present a dermal health hazard when tested as specified in 4.4.4.

#### 3.3.4 Color.

3.3.4.1 Class 1. The cloth shall be dyed and printed with the warp effect side as the face. The cloth shall be dyed to a ground shade approximating Khaki 475. The Woodland camouflage MARPAT shall be obtained by roller or screen-printing using either three or four rollers or screens, as appropriate for the Green 474, Khaki 475, Coyote 476 and Black 477 areas of the pattern with embedded EGA. Resin bonded pigments are not permitted except for a small amount of carbon black pigment may be used to meet the black shade, providing all other requirements are met.

3.3.4.2 Class 2. The cloth shall be dyed and printed with the warp effect side as the face. The cloth shall be dyed to a ground shade approximating Light Tan 479. The Desert camouflage MARPAT shall be obtained by roller or screen-printing using either three or four rollers or screens, as appropriate for the Urban Tan 478, Light Tan 479, Highland 480 and Light Coyote 481 areas of the pattern with embedded EGA. Resin bonded pigments are not permitted.

3.3.4.3 Class 3. The cloth shall be dyed and printed with the warp effect side as the face. The cloth shall be dyed to NWU II as specified in “Commercial US Governmentally Controlled Performance Specification Camouflage Print Performance Specification for AOR 1, AOR 2, NWU II, and NWU III.”

3.3.4.4 Class 4. The cloth shall be dyed and printed with the warp effect side as the face. The cloth shall be dyed to NWU III as specified in “Commercial US Governmentally Controlled Performance Specification Camouflage Print Performance Specification for AOR 1, AOR 2, NWU II, and NWU III.”

3.3.4.5 Knit cloth. The knit cloth shall be dyed Coyote 498.



3.3.5 Labile sulfur. If the use of dyes and compounds containing sulfur capable of oxidation to sulfuric acid is chosen, then they shall be applied such that the dyed cloth shall contain no more labile sulfur than shown by the standard sample when tested as specified in 4.4.5. When no standard sample is available, the dyed cloth shall show no more than a slight trace of labile sulfur when tested as specified in 4.4.5.

3.3.6 Color matching.

3.3.6.1 Visual matching. The color and appearance of the camouflage printed permethrin treated woven cloth, dyed permethrin treated knit cloth, and permethrin treated garments shall match the standard sample when tested in accordance with 4.4.5 and shall be a good match to the standard sample under horizon lamplight at  $2300 \pm 200$  K.

3.3.6.2 Instrumental color matching. Instrumental color matching is used as a tool to quantify shade evaluation if visually shade is rated unacceptable. All the colors in the Woodland MARPAT shall be instrumentally measured except for Khaki and all the colors in the Desert MARPAT shall be measured except for Highland given the areas of these exempted colors are too small for accurate instrumental readings. Each measured color shall match the standard sample. See 4.4.5.2.2 for evaluation procedure and acceptance requirements.

3.3.7 Colorfastness. When tested for the characteristics cited in table I, the colorfastness of the cloth shall be equal to or better than the standard sample, or equal to or better than the ratings specified in tables I and II.

3.3.8 Pattern execution.

3.3.8.1 Pattern execution (classes 1 and 2). The pattern for woodland and desert MARPAT shall reproduce the standard sample in respect to design, colors, and registration of the respective areas. The pattern repeat of the camouflage printed finished cloth shall be  $35 \pm 1\frac{1}{2}$  inches. Each pattern area shall show solid coverage; skitteriness exceeding that shown on the standard sample in any of the printed areas will not be acceptable. When the standard sample is not referenced for pattern execution, a pattern drawing will be provided, and the pattern of the finished cloth shall match that of the drawing.

3.3.8.2 Pattern execution (classes 3 and 4). The pattern for NWU II and NWU III shall be executed as specified in "Commercial US Governmentally Controlled Performance Specification Camouflage Print Performance Specification for AOR 1, AOR 2, NWU II, and NWU III."

3.3.9 Spectral reflectance.

3.3.9.1 Class 1 cloth. The finished cloth shall meet the spectral reflectance values (in percent) for the visible/near infrared wavelength range, 600 to 860 nanometers (nm) for the colors specified in table III as applicable when tested as specified in 4.4.5.

TABLE III. Class 1 cloth Spectral Reflectance Requirements.

| Wavelengths<br>Nanometers | Reflectance Values (Percent) |         |                           |         |              |         |
|---------------------------|------------------------------|---------|---------------------------|---------|--------------|---------|
|                           | Black<br>477                 |         | Coyote 476 &<br>Khaki 475 |         | Green<br>474 |         |
|                           | Minimum                      | Maximum | Minimum                   | Maximum | Minimum      | Maximum |
| 600                       | --                           | 10      | 8                         | 18      | 3            | 10      |
| 620                       | --                           | 10      | 8                         | 18      | 3            | 10      |
| 640                       | --                           | 10      | 8                         | 18      | 3            | 10      |
| 660                       | --                           | 10      | 8                         | 18      | 3            | 12      |
| 680                       | --                           | 10      | 10                        | 22      | 3            | 14      |
| 700                       | --                           | 10      | 18                        | 33      | 5            | 18      |
| 720                       | --                           | 10      | 22                        | 45      | 7            | 20      |
| 740                       | --                           | 10      | 30                        | 55      | 12           | 28      |
| 760                       | --                           | 10      | 35                        | 65      | 16           | 36      |
| 780                       | --                           | 10      | 40                        | 75      | 23           | 44      |
| 800                       | --                           | 10      | 45                        | 80      | 30           | 52      |
| 820                       | --                           | 10      | 50                        | 86      | 37           | 60      |
| 840                       | --                           | 10      | 55                        | 88      | 44           | 68      |
| 860                       | --                           | 10      | 60                        | 90      | 51           | 74      |

3.3.9.2 Class 2 cloth. The finished cloth shall meet the spectral reflectance values for the visible/near infrared wavelength range, 700 to 860 nanometers (nm) for the colors specified in table IV, when tested as specified in 4.4.5.

TABLE IV. Class 2 cloth Spectral reflectance requirements.

| Wavelengths<br>Nanometers | Reflectance values (percent) |         |                                  |         |                  |         |
|---------------------------|------------------------------|---------|----------------------------------|---------|------------------|---------|
|                           | Lt. Tan<br>479               |         | Lt. Coyote 481 &<br>Highland 480 |         | Urban Tan<br>478 |         |
|                           | Minimum                      | Maximum | Minimum                          | Maximum | Minimum          | Maximum |
| 700                       | 38                           | 53      | 19                               | 41      | 25               | 44      |
| 720                       | 38                           | 54      | 20                               | 41      | 25               | 45      |
| 740                       | 39                           | 55      | 20                               | 42      | 25               | 46      |
| 760                       | 40                           | 56      | 21                               | 42      | 26               | 47      |
| 780                       | 41                           | 57      | 21                               | 42      | 27               | 48      |
| 800                       | 43                           | 58      | 22                               | 43      | 28               | 50      |
| 820                       | 45                           | 59      | 23                               | 45      | 30               | 52      |
| 840                       | 48                           | 62      | 24                               | 46      | 33               | 55      |
| 860                       | 50                           | 65      | 25                               | 48      | 36               | 58      |

3.3.9.3 Class 3 and 4. The finished class 3 and 4 cloth shall meet the spectral reflectance values (in percent) for the visible/near infrared wavelength range, for the colors specified in “Commercial US Governmentally Controlled Performance Specification Camouflage Print Performance Specification for AOR 1, AOR 2, NWU II, and NWU III.”

3.3.9.4 Knit cloth. The finished cloth shall meet the spectral reflectance values for the visible/near infrared wavelength range, 600 to 860 nanometers (nm) for Coyote 498 specified in table V, when tested as specified in 4.4.5.1.

TABLE V. Knit cloth Spectral reflectance requirements.

| Reflectance values (percent) |            |         |
|------------------------------|------------|---------|
| Wavelengths<br>Nanometers    | Coyote 498 |         |
|                              | Minimum    | Maximum |
| 600                          | 8          | 23      |
| 620                          | 8          | 23      |
| 640                          | 8          | 25      |
| 660                          | 8          | 33      |
| 680                          | 10         | 45      |
| 700                          | 18         | 56      |
| 720                          | 22         | 63      |
| 740                          | 30         | 68      |
| 760                          | 35         | 71      |
| 780                          | 40         | 72      |
| 800                          | 45         | 75      |
| 820                          | 50         | 77      |
| 840                          | 55         | 77      |
| 860                          | 60         | 77      |

### 3.3.10 Dimensional stability.

3.3.10.1 Dimensional stability (class 1-4 basic material). The shrinkage in warp and filling direction of the untreated finished cloth shall be not greater than 5.0 percent for individual sample unit and not greater than 5.0 percent for the lot average when tested as specified in 3.3.1 and 4.4.5. The fabric shall not elongate more than 1.0 percent. Manufacturers must compensate for actual fabric shrinkage in order to deliver finished, treated garments complying with the dimension requirements specified in 4.4.5.

3.3.10.2 Dimensional stability (knit cloth). The shrinkage in course and wale direction of the untreated finished cloth shall be not greater than 10.0 percent for individual sample unit and not greater than 8.0 percent for the lot average when tested as specified in 3.3.1 and 4.4.5. Manufacturers must compensate for actual fabric shrinkage in order to deliver finished, treated garments complying with the dimension requirements specified in 4.4.5.

3.3.11 Permethrin Content and Insect Bite Protection. The finished garments with permethrin treatment shall be strictly limited to the level specified in 3.3.11.1 and provide the minimum insect bite protection specified in 3.3.11.2 when tested as specified in 4.4.5. with Federal Insecticide, Fungicide and Rodenticide Act (FIFRA) as amended (see 2.2.2).

3.3.11.1 Permethrin Treatment. The cloth shall be treated with an insect repellent treatment conforming to the requirements of MIL-PRF-32659, except that the active ingredient concentration and mosquito percent bite protection shall conform to the values below when tested as specified in 4.4.5. The permethrin treatment shall be durable to repeated launderings to provide the minimum permethrin level after 20 launderings as specified below. The finished permethrin treatment shall not degrade any performance characteristics or present any latent defects to the cloth or garment. Permethrin content, expressed in milligrams per square centimeter, shall be measured prior to laundering and after 20 laundering cycles.

INITIAL

|                                      | Minimum | Maximum |
|--------------------------------------|---------|---------|
| Blouse woven fabric (see 3.3.1)      | 0.102   | 0.138   |
| Trousers woven fabric (see 3.3.1)    | 0.102   | 0.138   |
| Blouse torso knit fabric (see 3.3.2) | 0.084   | 0.122   |

AFTER 20 Launderings

|                                      |       |
|--------------------------------------|-------|
| Trousers and Blouse sleeves & collar | 0.042 |
| Blouse torso knit fabric             | 0.051 |

3.3.11.2 Percent Bite Protection. Finished permethrin treated trousers shall provide bite protection specified below when assessed by the bite protection testing specified in 4.4.6.6. Government approval is required initially, and any time there is a change in the permethrin treatment formulation or processing conditions (see 4.4.5.5).

| Condition            | Percent Bite Protection |
|----------------------|-------------------------|
| Initial              | ≥ 85 percent            |
| After 20 launderings | ≥ 80 percent            |
| After 50 launderings | ≥ 75 percent            |

3.3.12 Ground shade/printed seconds/dyed seconds/mill seconds. Ground shade cloth shall be dyed in conformance with the specified basic material and shall meet the physical, mechanical, and dimensional requirements of the respective finished cloth. Printed seconds shall be defined as cloth that has been rejected only for defects pertaining to color, infrared reflectance, or camouflage print patterns, which are cited in the specified basic material requirements. Dyed seconds shall be defined as cloth that has been rejected only for defects pertaining to color or infrared reflectance, which are cited in the specified basic material requirements. Mill seconds shall be cloth that has been rejected for visual defects only, and dyed to match ground shade (see 3.5.3). Mill seconds finished firsts may contain slubs and knots (see 4.4.6 and ASTM D3990 for all other cloth defects which constitute seconds).

3.3.13 Disposal of ground shade/printed seconds/dyed seconds/mill seconds/rejected garments. All scraps, irregulars, extra material, and garments containing intellectual property/trademarks which are not utilized for Government contracts or a purpose authorized in writing by the Government, shall be destroyed and not sold or transferred in any manner. This restriction applies to the prime contractor, as well as all subcontractors. Contractor shall be prepared to certify as to the method and execution of the destruction of all scraps, seconds, irregulars, extra material, and garments containing the aforementioned intellectual property/trademarks.

3.4 System level garment requirements. The finished garments shall conform to the requirements in table VI when tested as specified in 4.4.5.

TABLE VI. System Testing Requirements for Garment.

| CHARACTERISTIC                                                                                   | REQUIREMENT  |
|--------------------------------------------------------------------------------------------------|--------------|
| pH                                                                                               | 5.0 - 8.5    |
| Elastic Force vs. Elongation<br>Cargo Pocket<br>Waistband                                        | See 3.6.3.   |
| Seam Strength (minimum)                                                                          | See 3.4.2    |
| Permethrin Content                                                                               | See 3.3.11   |
| Bite Protection, percent                                                                         | See 3.3.11.2 |
| Flammability (After 1 and 25 laundering cycles)<br>Instrumented Manikin Test (4 second exposure) | See 3.4.1    |

3.4.1 Instrumented Manikin Test. When tested in accordance with 4.4.5, the EFRCE shall provide FR flash fire protection and shall have no more than 25 percent burn injury prediction including second and third degree during a four second exposure at a 2.0 calorie per centimeter squared per second heat flux.

3.4.2 Seam Strength. Finished garments shall conform to the following minimum seam strength requirements when tested in accordance with 4.4.5. For seams joining woven cloth to woven cloth, the minimum seam strength shall be 90 pounds. For seams joining woven cloth to knit cloth, the minimum seam strength shall be 30 pounds. For seams joining knit cloth to knit cloth, the minimum seam strength shall be 30 pounds.

3.4.3 Antibacterial properties testing. The knit cloth shall contain odor reducing properties showing a minimum of 90 percent reduction of bacteria when tested against staphylococcus aureus and pseudomonas aeruginosa bacteria when tested in accordance with 4.4.5 with clarifications specified in 4.4.5.9.

### 3.5 Design.

3.5.1 Blouse. The blouse shall be a multi-cloth design using two separate cloths for the main garment construction; a knit cloth for the torso and a woven cloth for the sleeves and collar. The blouse uses a pullover shirt design with long sleeves. Both sleeve and collar are constructed of woven cloth; the torso is a knit cloth. The collar converts to a mock turtle neck for additional protection in a flame environment. The slide fastener front opening extends from the top of collar to the mid-chest, with a padded placket beneath the slide fastener, which improves comfort when the blouse is worn under the ballistic vest. There are two upper sleeve pockets with bottom bellows, a drainage hole located in the bottom bellow of each pocket, with a tapered flap and concealed two-button closure. Sleeves are one-piece, single layer with adjustable wrist tabs at sleeve hem for wrist closure adjustment. There are adjustable tabs at wrists having a single button located on the underside of the sleeve tab and three buttonholes in the sleeve hem for wrist closure adjustment. The buttonholes shall be eyelet end, tapered bar, with 7/8 inch cut opening and 52–56 stitches per inch. The buttonhole purling shall be between tab (see Figure 8). The buttonhole purling shall be between tab, flap plies and cuff on the outside of the blouse. The front cut edge of buttonholes shall be positioned  $3/4 \pm 1/8$  inch from front folded edge for the blouse. The pocket drainage eyelets shall have a minimum of 19 stitches each, with the purling on the outside. Material edges must not ravel. The blouse has external single layer reinforcement patches at the elbows.

3.5.2 Trousers. The trousers have a concealed button and buttonhole fly, front panels with a centered single stitched pleat (the fold of the pleat faces the outseam), hemmed legs, seven evenly-spaced  $1 \pm 1/8$  inch wide belt loops with a 3-inch belt loop opening and side waist elastic adjustments. The buttonholes shall be eyelet end, tapered bar, with  $7/8$  inch cut opening and 52–56 stitches per inch. The elasticized portion of the waist starts at the front of the pocket bearer and extends past the outseam onto the back of the trousers to achieve the finished four inches of the waist circumference for each size. The trousers have two side hanging pockets and two welt-opening hip pockets. There are two pleated cargo type pockets (the pleats face the back of the trousers). The larger pocket has a flap that is bartacked at forward lower corner and one rear button closure, a bottom bellow and elasticized pocket upper edge creating an adjustable opening to secure the contents when the flap is not buttoned. The elastic cargo pocket edge requires elongation/extension when the hand is inserted into the pocket yet does not distort the trouser leg when lying in the gathered/relaxed closed position. The smaller cargo pocket has a flap with two button closures and is not elasticized. Drainage holes are positioned in the cargo pocket bellow (positioned near the back of the pocket). There are two calf pockets (the pleats face the back of the trousers) with a flap bartacked at forward lower corner and a two button closure, to secure the contents when the flap is buttoned. The calf pocket must not distort the trouser leg when lying in the relaxed closed position. All pockets, except for the side hanging ones have a flap with fusible interlining and closure with concealed buttons. The trousers have single layer external reinforcement patches at the seat and knees. The buttonholes shall be eyelet end, tapered bar, with  $7/8$  inch cut opening and 52 – 56 stitches per inch. The buttonhole purling shall be between tab and flap plies, front facing and front plies, fly and front waist, and on the outside of the trouser. The front cut edge of buttonholes shall be positioned  $1/2 \pm 1/16$  inch from front folded edge for the trouser. The pocket drainage eyelets shall have a minimum of 19 stitches each, with the purling on the outside. Material edges must not ravel.

### 3.6 Components.

3.6.1 Thread. The thread for the bobbin/looper/needle shall be aramid, Tex Size 40 conforming to A-A-55217 or as an alternate, para-aramid spun staple thread, Tex Size 40 conforming to A-A-55195, Type I, may be used.

3.6.1.1 Thread colorfastness and color. All thread shall be non-staining and show good colorfastness to laundering when tested as specified in 4.4.2. The thread color shall be Khaki P-1 for classes 2 and 3 and Camouflage Green 483 for classes 1 and 4..

3.6.2 Button Attachment Thread. All buttons shall be attached using aramid, Tex Size 40 conforming to A-A-55217.

3.6.3 Elastic Webbing. The elastic webbing for the trouser waistband and cargo pockets shall be 1 inch wide and conform to performance of Type II, class 1 of MIL-W-5664D or QST style 150-43171RA,  $1 \pm 1/16$  inches or equal. Elastic ends shall be heat cut to prevent raveling. The color shall be natural for either class. The elastic for the waistband shall be cut to the length needed to achieve both the trouser half waist finished relaxed and finished stretched measurements (for each size) as specified in paragraph 3.10, and the elongation force specified in table VII. The elastic in the cargo pocket opening shall extend the full width to achieve a 6-3/4 to 7-1/2 inch relaxed opening. The elastic will retreat to its relaxed opening position when the hand is removed. The encased elastic webbing for the cargo pocket shall lie flat in the relaxed state and not distort the trouser leg. The cargo pocket opening shall retain its elongation/recovery characteristic for the life of the garment. The finished elasticized waistband and cargo pocket features shall conform to the requirements listed in table VII.

TABLE VII. Elastic Elongation.

| Feature      | Elongation (inches)<br>±1/16 | Force (pounds) Minimum to<br>Maximum (inclusive) | Test Method |
|--------------|------------------------------|--------------------------------------------------|-------------|
| Cargo Pocket | 1-1/2                        | 2.5 to 3.3                                       | See 4.4.5.4 |
| Waistband    | 3/4                          | 0.5 to 1.5                                       | See 4.4.5.4 |
| Waistband    | 1-1/2                        | 1.5 to 4.0                                       | See 4.4.5.4 |

3.6.4 Fastener Tape, Loop, IR Tab. The loop fastener tapes for the left and right sleeve pocket flap IR tab shall be 2 inches wide and 2 inches long, CG 483 for classes 1 and 4 and Khaki P-1 C.A.66019 for classes 2 and 3 and shall conform to type II, class 1 of A-A-55126. Cut edges shall be finished so that they do not ravel.

3.6.5 Zippers. The blouse front zipper shall finish 8-1/2 inches long and shall conform to A-A-55634, and shall be Coyote 498 in color. Zipper shall be Type I, Style 4, Size 3 continuous-element fastener according to A-A-55634. Zipper stop shall finish at the base of the finished zipper opening.

3.6.6 Foam. The flame resistant (FR) foam padding stitched into the zipper flap shall conform to MIL-PRF-26514; Type 1, class 2, grade C, 1/4 inch thick.

3.6.7 Interlining. To improve the abrasion resistance and appearance the pocket flaps shall be interlined. The fusible interlining shall have the same life expectancy as the blouse, and after attaching to the basic cloth, shall meet an initial directional bond strength of 32 ounces minimum per inch and 24 ounces minimum per inch after 3 launderings when tested as specified in 4.4.1. There shall be no delamination, bubbling or strike-through after a blouse is laundered three times.

3.6.8 Buttons. The buttons shall be dull finish, 30-ligne, in accordance with A-A-59994, Style 26. The color of the buttons shall be a good match to Coyote 498. When attached to the blouse and the trouser, the button and thread shall withstand a pull test of 40 lb. (min.), when tested as specified in 4.4.3.3.

### 3.7 Labels.

3.7.1 Blouse labels. The blouse shall contain 3 permanent labels as specified below. All permanent label inscription, legibility, label material, and label attachment method shall be expected to last the expected life of the uniform.

3.7.1.1 Flame Resistant Organizational Gear (FROG) label. Each blouse shall have a woven FROG label, 63 mm x 63 mm, cut single, fused edge, in the following configuration. The woven FROG label shall be sewn on each blouse at the right lower front (as worn), to the left of the blouse combination label and 1-1/2 inches above the hem.



Garment Class 1 label



Garment Class 2 label

3.7.1.2 Blouse combination label. A separate combination label (see 6.7 for example) shall contain size, body measurement, identification, and care information as specified below. The information needed to designate size and body measurement in the top portion of the combination label is specified in the table VII “Size and Body Measurement.” The label color shall be Coyote 498. The inscription shall have a minimum font size of 10 points. The inscription legibility, label, and label attachment method shall last the expected life of the uniform. The combination label shall be sewn on the right lower front (as worn), 3/4 inch to the left of the side seam and 1-1/2 inch above the bottom hem. Label size shall be no longer than 4 inches and no wider than 2-1/2 inches. The size – length designation (with abbreviation), body measurements (see 3.6.3), and stock number shall be included on the combination label. Labelling shall include the following identifying information. The class on line 1 and the PATTERN on line 2 shall correspond to blouse class ordered and delivered.

Type I, Class CLASS Enhanced Flame Resistant Combat Ensemble (EFRCE) Blouse-  
PATTERN, with Durable Insect Protection

Woven Cloth fiber content

Knit Cloth fiber content

All classes shall have the following information below the identifying information:

Knit cloth fiber content

Contract Information:

Contract Number

Contract Name

3.7.1.3 Care information. For all classes, a solid line shall be placed below the identifying information, and the following care information shall be located below the solid line on the label.

Care Information:

**DO NOT BLEACH, STARCH, DRY CLEAN, OR PRESS. DO NOT USE DRYER SHEETS**

1. Washing. Machine wash using Permanent Press Cycle or hand wash in warm water using mild detergent that does NOT contain optical brighteners. Rinse completely. Do not overload the machine.

2. Drying. Tumble dry on low heat. Do not overload the dryer.

3. Fabric softener. The use of fabric softeners is not recommended due to potential to adversely affect the flame protection.

3.7.1.4 Blouse insect protection label. Finished blouses shall include a permanent insect protection label complying with an approved EPA registration (see 6.9). The label shall be heat sealed or stitched on all four sides and attached to left lower front (as worn). Labeling information, characteristics and location of labels are subject to Government approval prior to production. See paragraph 3.7.5 for additional EPA hang tag information requirement.

3.7.1.5 Blouse insect protection brand label. Finished blouses shall contain brand identification in accordance with the treatment’s EPA registration. Brand identification may be included in an expanded permanent insect protection label (see 3.4.3.1.3) or as a separate brand label stitched directly above the insect protection label. The contractors labeling is subject to Government approval prior to production.



3.7.2 Trouser labels. The pair of trousers shall contain 3 permanent labels as specified below. All permanent label inscription, legibility, label material, and label attachment method shall last the expected life of the uniform.

3.7.2.1 FROG label. Each pair of trousers shall have a woven FROG label, 63 mm x 63 mm, cut single, fused edge, in the configuration shown in 3.7.1.1. A woven FROG label shall be sewn on each pair of trousers on the left back pocket (as worn) so that on the finished trousers the label shall face the wearer. The FROG label shall be positioned to the left (as worn) of the trouser combination label.

3.7.2.2 Trouser combination label. A separate combination label (see 6.8 for example) shall contain size, body measurement, identification, and care information as specified below. The information needed to designate size and body measurement in the top portion of the combination label is specified in the table VII "Size and Body Measurement." The label color shall be Coyote 498. The inscription shall have a minimum font size of 10 points. The inscription legibility, label, and label attachment method shall last the expected life of the uniform. The combination label shall be stitched on all four sides and attached to the underside of the left back pocket with the writing facing out as worn, parallel to the FROG label. The attachment of this label will not be visible on the outside of the garment. Label size shall be no longer than 4 inches and no wider than 2-1/2 inches. The size – length designation (with abbreviation), body measurements (see 3.6.3), and stock number shall be included on the combination label. Labelling shall include the following identifying information. The CLASS on line 1 and the PATTERN on line 2 shall correspond to trouser class ordered and delivered.

Type I, Class CLASS Enhanced Flame Resistant Combat Ensemble (EFRCE) Trouser-  
PATTERN, with Durable Insect Protection  
Woven Cloth fiber content

Contract Information:

Contract Number

Contract Name

3.7.2.3 Care information. The care information in 3.7.1.3 shall appear below a solid line.

3.7.2.4 Trouser insect protection label. Finished trousers shall include a permanent insect protection label complying with an approved EPA registration (see 6.9). The label shall be heat sealed or be stitched on all four sides and attached to the underside of the right back pocket (see Figure 6) with the writing facing the wearer. The attachment of this label will not be visible on the outside of the garment. Labeling information, characteristics and location of labels are subject to Government approval prior to production. See paragraph 3.7.5 for additional EPA hang tag information requirement.

3.7.2.5 Trouser insect protection brand label. Each trouser shall contain brand identification in accordance with the treatment's EPA registration. Brand identification may be include in the permanent insect protection label (see 3.7.2.4) or as a separate label stitched directly above the insect protection label which shall not be visible on the outside of the garment. The contractors labeling is subject to Government approval prior to production (see Figure 6).

3.7.3 Size and body measurements table.TABLE VIII. Size and Body Measurements.

| Location                       | Size     | Measurement             |
|--------------------------------|----------|-------------------------|
| Chest<br>(Size of Blouse)      | X-Small  | Up to 33 inches         |
|                                | Small    | 33 to 37 inches         |
|                                | Medium   | 37 to 41 inches         |
|                                | Large    | 41 to 45 inches         |
|                                | X-Large  | 45 to 49 inches         |
|                                | XX-Large | 49 to 53 inches         |
| Height<br>(Length of Blouse)   | XX-Short | 55 to 59 inches         |
|                                | X-Short  | 59 to 63 inches         |
|                                | Short    | 63 to 67 inches         |
|                                | Regular  | 67 to 71 inches         |
|                                | Long     | 71 to 75 inches         |
|                                | X-Long   | 75 to 79 inches         |
|                                | XX-Long  | Above 79 inches         |
| Waist<br>(Size of Trouser)     | X-Small  | Up to 27 inches         |
|                                | Small    | 27 to 31 inches         |
|                                | Medium   | 31 to 35 inches         |
|                                | Large    | 35 to 39 inches         |
|                                | X-Large  | 39 to 43 inches         |
|                                | XX-Large | Over 43 inches          |
| Inseam<br>(Length of Trousers) | X-Short  | Up to 26-1/2 inches     |
|                                | Short    | 26-1/2 to 29-1/2 inches |
|                                | Regular  | 29-1/2 to 32-1/2 inches |
|                                | Long     | 32-1/2 to 35-1/2 inches |
|                                | X-Long   | 35-1/2 to 38-1/2 inches |
|                                | XX-Long  | Over 38-1/2 inches      |

3.7.4 Garment Lot Designation. For garment manufacturing traceability, each blouse and trouser garment shall have a lot designation in accordance with lot Numbering procedure as specified in DLA Troop Support Clothing & Textiles Additional Quality Assurance Requirements. The lot number for the blouse shall be placed inside the right sleeve behind the elbow patch reinforcement. The lot number for the trouser shall be placed next to or above the combination identification/care label on the trouser. Batch designation is not acceptable.

3.7.5 Hang Tag Labels.

3.7.5.1 Bar Code Label. Each blouse and trouser shall have an individual bar code placed on a paper tag for personal clothing items. The paper tag shall be standard bleached sulfate having a basis weight of 100 pounds. The paper used for the tags shall have a smooth finish to accept thermal transfer and direct printing. The tags shall have a hole and shall be attached to each item by a fastener. The tags shall be clearly legible and readable by a scanner. such as letters and numbers that can be read by people, as opposed to symbol characters within bar codes or electronic data within tags, which are read by machines. The bar-coding element shall be a 13 digit national stock number (NSN). The bar-code for NSN printing shall be a medium to high code density and shall be located so that it is completely visible on the item when it is folded and/or packaged as specified and shall cause no damage to the item. The barcode shall

be printed with a human-readable interpretation (HRI). The NSN must also be placed on all shipping cartons.

3.7.5.2 Insect protection label. Each garment shall have an individual paper tag attached that provides additional insect protection information required by EPA registration and subject to Government approval (see Figure 7).

3.8 List of pattern parts. The Government shall furnish a complete set of patterns which show directional line markings for proper assembly. The Government patterns are to be used as a guide for cutting contractor's working patterns. Minor modifications are permitted to accommodate manufacturing procedures; however, the design, finished appearance and finished measurements must be maintained. Standard patterns provide a seam allowance of 3/8 inch for single needle seams and 1/2 inch for double needle seams. Buttonholes, pockets and pocket flaps shall be located in accordance with marks on patterns and figures. The pattern parts list in tables IX and X are provided for first quality fabrics and to ensure that the pattern set is complete.

TABLE IX. List of blouse pattern parts.

| Material    | Nomenclature              |
|-------------|---------------------------|
| Woven Cloth | Front Yoke                |
|             | Sleeve                    |
|             | Collar                    |
|             | Placket                   |
|             | Zipper Flap               |
|             | Sleeve Pocket             |
|             | Sleeve Pocket Flap Tab    |
|             | Elbow Patch               |
|             | Cuff Tab                  |
|             | Lower Sleeve Pocket       |
| Knit Cloth  | Front                     |
|             | Back                      |
|             | Sleeve Gusset             |
| FR FOAM     | Zipper Flap Reinforcement |

TABLE X. List of trouser pattern parts.

| Pattern Abbreviation | Nomenclature                |
|----------------------|-----------------------------|
| Woven Cloth          | Front                       |
|                      | Back                        |
|                      | Cargo Pocket                |
|                      | Cargo Pocket Facing         |
|                      | Cargo Pocket Flap Tab       |
|                      | Thigh Cargo Pocket          |
|                      | Thigh Cargo Pocket Flap Tab |
|                      | Side Hanging Pocket         |
|                      | Side Hanging Pocket Bearer  |
|                      | Side Hanging Pocket Facing  |
|                      | Hip Pocket                  |
|                      | Hip Pocket Bearer           |
|                      | Hip Pocket Facing           |
|                      | Hip Pocket Flap Tab         |
|                      | Lower Leg Pocket            |
|                      | Lower Leg Pocket Flap       |
|                      | Gusset                      |
|                      | Left Fly                    |
|                      | Right Fly Lining            |
|                      | Seat Patch                  |
|                      | Knee Patch                  |

3.8.1 Parts cut from ends and from ground shade. Use of ground shade and/or print second quality is restricted to: hip pocket, side hanging pocket, cargo pocket facing pattern parts and waistband lining. Printed seconds shall be defined as cloth that has been rejected only for defects pertaining to color, infrared reflectance, or camouflage print patterns, which are cited in the specified basic material requirements. No portion of any part cut from these materials will be visible in the finished garment.

3.9 Configuration. The following specifications are needed to provide uniform appearance, comfort and durability in combat operations. End item blouse and trouser construction and appearance shall conform to the requirements of this document and the finished dimensions in tables XIV and XV (see 4.4.8) and figures 1 – 7 to maintain item configuration and compliance to component and end item tests (see 4.4).

3.9.1 Seaming and stitching. Seaming shall be consistent, exhibit a uniform appearance and shall conform to the ASTM D6193 stitch types listed in table XI. The backside of all seams (inside garment) shall be flat with no protruding seam allowance or raw edges to create irritation, discomfort or poor appearance. To maintain durability and functionality, the seams shall be sewn with 10-14 stitches per inch for all outside visible stitching. Thread ends shall be secured by a minimum of 1/4 inch backstitched or 1/2-inch overlap when broken stitches are repaired. Overedge or pre-hemming shall be 6 to 10 stitches per inch. Bartacks shall be in accordance with table XII. Bartacks are required at the specified reinforcement points (see figures 1-4) and centered on belt loop for attachment. The pocket drainage eyelets shall have a minimum of 19 stitches each, with the purling on the outside. Material edges must not ravel. Raw edges may be turned-in, turned- under or serged to prevent raveling.

TABLE XI. Seam Types.

| Seam Placement                                                                        | Seam Type                   | Gage                                                           | Stitch Type             |
|---------------------------------------------------------------------------------------|-----------------------------|----------------------------------------------------------------|-------------------------|
| Blouse sleeve/side seams                                                              | SSa-2                       | 3/16 to 9/32 inch                                              | 301 or 401 and 504      |
| Shoulder/underarm seams, yoke seams                                                   | LSbm-3                      | 3/16-1/4 inch                                                  | 301 or 401 and 504      |
| Outseams, seat seam and inseams                                                       | LSc-2                       | 3/16 to 9/32 inch gage                                         | 301 or 401 and 504      |
| Waistband                                                                             | LSk-2                       | 1-1/2 inch                                                     | 301 or 401              |
| Alternate Waistband- metered elastic                                                  | LSbc-4                      | 1/                                                             | 1/                      |
| Top stitching for pocket flaps, collar, cuff tab, flies, tabs, and front edges.       | SSe-1                       | 1/8 to 1/4 inch from the edge (uniform throughout the garment) | 301                     |
| Closing of side hanging and hip pockets leaving 1/2 inch chain                        | SSe-2 and OSf-1<br>Or SSa-2 | 3/16 to 1/4 inch from edges                                    | 301 or 515, 516, or 519 |
| Hip pocket                                                                            | SSa or SSbe-2               | Hip pocket welt width 1/4 - 3/8 inch:                          | 301                     |
| Attachment of patch pockets and attachment of side and hip pocket bearers and facings | LSd-1                       | one row 1/16 to 1/8 inch apart                                 | 301                     |
| Double needle flap stitching 2/                                                       | LSd-2                       | two rows 3/16 to 1/4 inch apart                                | 301                     |
| Belt loop construction                                                                | Efn-2 or Efp-2              | 1/16 to 1/8 inch from edges                                    | 301                     |
| Attachment of elbow, seat and knee patches                                            | LSd-2                       | two rows 3/16 to 1/4 inch apart                                | 301                     |
| Label attachment                                                                      | LSbj-1                      | 1/8 to 3/16 inch from edges                                    | 301                     |
| All hems: Trouser legs and Sleeves                                                    | EFb-1                       | 1/16 to 1/8 inch from edge                                     | 301                     |
| Blouse hem                                                                            | EFa-2                       | 1/4 to 3/8 inch gage                                           | 602                     |
| Internal pocket bags                                                                  | EFd-1                       | 1/4- 3/16 from raw edge                                        | 503 or 504              |
| Finish Belt Loops                                                                     | EFa-1 or heat seal          | 1/8-1/4 from raw edge                                          | 401                     |

1/ An alternate waistband construction shall use 1-1/2  $\pm$  1/16 inch elastic metered its entire length using seam type LSbc-4 with the elastic inserted within the seam construction along opposite sides of the waistband per pattern placement. A total of 4 stitch lines shall be correctly tensioned or balanced such that the stitch lines do not impede stretch whereby stitches could break open. The alternate waistband shall meet elastic webbing, elongation, and force specified in 3.4.2.3 and finished dimensions specified in 4.4.7

2/ An alternate for all pocket flaps shall be a single needle stitch attachment in accordance with the pattern then turned and topstitched 3/16 inch to 1/4 inch with backtacking at each end. The top flap edge, raw or overedged, shall not be visible when the flap is raised.

TABLE XII. Bartacks.

| Location                                                                     | Size (Inches) | Stitches/bartack |
|------------------------------------------------------------------------------|---------------|------------------|
| BLOUSE                                                                       |               |                  |
| Top of slide fastener                                                        | 3/8           | 20-27            |
| Top corners of sleeve pockets                                                | 3/8           | 20-27            |
| Sleeve pocket flaps                                                          | 3/8           | 20-27            |
| Top corners of pen pocket                                                    | 3/8           | 20-27            |
| Top of pen pocket divider                                                    | 3/8           | 20-27            |
| Vertically at each end of cuff tab along vertical stitching                  | 3/8           | 20-27            |
| End of each sleeve inseam at cuff edge                                       | 3/8           | 20-27            |
| TROUSER                                                                      |               |                  |
| Across top and bottom of belt loops                                          | 1             | 52-56            |
| Each end of top front/hip pocket opening                                     | 5/8           | 35-40            |
| Each end of back hip pocket welt openings                                    | 5/8           | 35-40            |
| Top corners of cargo, thigh, and calf pockets                                | 1/2           | 28-34            |
| Top corners of cargo, thigh, and calf pocket flaps                           | 1/2           | 28-34            |
| Bottom front corner of cargo pocket flap                                     | 1/2           | 28-34            |
| Bottom corners of cargo and thigh pocket and bottom at shared seam           | 1/2           | 28-34            |
| Across pleat placements below lower edge of elastic on top of cargo pockets. | 3/8           | 20-27            |
| Across pleat placements on top of thigh and calf pockets                     | 3/8           | 20-27            |
| Bottom of fly, Superimposed on "J" stitching                                 | 3/8           | 20-27            |
| Top and bottom of crotch gusset                                              | 3/8           | 20-27            |
| Ends of waistband and cargo pocket elastic                                   | 1-1/8         | 55-65            |

3.10 Finished dimensions (all types and classes). The finished and treated blouse and trouser shall conform to the dimension listed in tables XIII and XIV. Finished items shall be laid flat on a table and allowed to rest prior to measurement.

TABLE XIII. Blouse Finished Measurement (inches).

| SIZE                    | XX-SHORT | X-SHORT | SHORT  | REGULAR | LONG   | X-LONG | XX-LONG | TOL.  |
|-------------------------|----------|---------|--------|---------|--------|--------|---------|-------|
| Half Chest -- 1/        |          |         |        |         |        |        |         |       |
| X-Small                 | -        | 19      | 19     | 19      | --     | --     | --      | ± 3/4 |
| Small                   | 21       | 21      | 21     | 21      | 21     | 21     | --      |       |
| Medium                  | 23       | 23      | 23     | 23      | 23     | 23     | 23      |       |
| Large                   | --       | 25      | 25     | 25      | 25     | 25     | 25      |       |
| X-Large                 | --       | --      | 27     | 27      | 27     | 27     | 27      |       |
| XX-Large                | --       | --      | --     | 29      | 29     | 29     | 29      |       |
| Half Bottom -- Width 2/ |          |         |        |         |        |        |         |       |
| X-Small                 | --       | 15-3/4  | 15-3/4 | 15-3/4  | --     | --     | --      | ± 3/4 |
| Small                   | 17-3/4   | 17-3/4  | 17-3/4 | 17-3/4  | 17-3/4 | 17-3/4 | --      |       |
| Medium                  | 19-3/4   | 19-3/4  | 19-3/4 | 19-3/4  | 19-3/4 | 19-3/4 | 19-3/4  |       |
| Large                   | --       | 21-3/4  | 21-3/4 | 21-3/4  | 21-3/4 | 21-3/4 | 21-3/4  |       |
| X-Large                 | --       | --      | 23-3/4 | 23-3/4  | 23-3/4 | 23-3/4 | 23-3/4  |       |
| XX-Large                | --       | --      | --     | 25-3/4  | 25-3/4 | 25-3/4 | 25-3/4  |       |

TABLE XIII. Blouse Finished Measurement (inches). Continued

| SIZE                               | XX-SHORT | X-SHORT | SHORT  | REGULAR | LONG   | X-LONG | XX-LONG | TOL.  |
|------------------------------------|----------|---------|--------|---------|--------|--------|---------|-------|
| Back Length -- 3/                  |          |         |        |         |        |        |         |       |
| X-Small                            | --       | 27-3/4  | 28-3/4 | 30-1/4  | --     | --     | --      | ±3/4  |
| Small                              | 27-1/4   | 28-1/4  | 29-1/4 | 30-3/4  | 32-1/8 | 33-1/2 | --      |       |
| Medium                             | 27-3/4   | 28-3/4  | 29-3/4 | 31-1/4  | 32-5/8 | 34     | 35-3/8  |       |
| Large                              | --       | 29-1/4  | 30-1/4 | 31-3/4  | 33-1/8 | 34-1/2 | 35-7/8  |       |
| X-Large                            | --       | --      | 30-3/4 | 32-1/4  | 33-5/8 | 35     | 36-3/8  |       |
| XX-Large                           | --       | --      | --     | 32-3/4  | 34-1/8 | 35-1/2 | 36-7/8  |       |
| Sleeve Length -- 4/                |          |         |        |         |        |        |         |       |
| X-Small                            | --       | 30      | 31     | 32      | --     | --     | --      | ± 3/4 |
| Small                              | 30-1/2   | 31-1/2  | 32-1/2 | 33-1/2  | 34-1/2 | 35-1/2 | --      |       |
| Medium                             | 32       | 33      | 34     | 35      | 36     | 37     | 38      |       |
| Large                              | --       | 34-1/2  | 35-1/2 | 36-1/2  | 37-1/2 | 38-1/2 | 39-1/2  |       |
| X-Large                            | --       | --      | 37     | 38      | 39     | 40     | 41      |       |
| XX-Large                           | --       | --      | --     | 39-1/2  | 40-1/2 | 41-1/2 | 42-1/2  |       |
| Half Sleeve Cuff -- 5/             |          |         |        |         |        |        |         |       |
| X-Small                            | --       | 6-1/2   | 6-1/2  | 6-1/2   | --     | --     | --      | ± 1/4 |
| Small                              | 6-7/8    | 6-7/8   | 6-7/8  | 6-7/8   | 6-7/8  | 6-7/8  | --      |       |
| Medium                             | 7-1/4    | 7-1/4   | 7-1/4  | 7-1/4   | 7-1/4  | 7-1/4  | 7-1/4   |       |
| Large                              | --       | 7-5/8   | 7-5/8  | 7-5/8   | 7-5/8  | 7-5/8  | 7-5/8   |       |
| X-Large                            | --       | --      |        |         |        |        |         |       |
| XX-Large                           | --       | --      |        |         |        |        |         |       |
| Collar Height at Center Back -- 6/ |          |         |        |         |        |        |         |       |
| All Sizes                          | 3-1/4    | 3-1/4   | 3-1/4  | 3-1/4   | 3-1/4  | 3-1/4  | 3-1/4   | ± 1/4 |
| Collar Height at Front Edge -- 7/  |          |         |        |         |        |        |         |       |
| All Sizes                          | 2-1/2    | 2-1/2   | 2-1/2  | 2-1/2   | 2-1/2  | 2-1/2  | 2-1/2   | ± 1/4 |

1/ Half Chest – With blouse zipped up, measurement from side seam folded edge to folded edge across blouse chest in line with pit of armhole (bottom of seam) (see Figure 9) .

2/ Half Bottom Width – Measurement from side seam folded edge to folded edge across blouse bottom at bottom edge.

3/ Back Length – Measurement along center back from collar seam to bottom edge of blouse (in line with grain of fabric for straight line).

4/ Sleeve Length – Measurement along folded edge of the top sleeve from the center back to the bottom of the sleeve cuff with sleeve folded along underarm seam, (see Figure 10).

5/ Half Cuff – Measurement at bottom of cuff along edge from folded edge to folded edge.

6/ Collar Height at Center Back – Measurement along center back of collar from setting seam to top of collar.

7/ Collar Height at Front Edge – Measurement along center front edge of collar from setting seam to top of collar.

TABLE XV. Trouser – Finished Measurements (inches)

| SIZE                      | X- SHORT | SHORT  | REGULAR | LONG   | X-LONG | XX- LONG | TOL. |
|---------------------------|----------|--------|---------|--------|--------|----------|------|
| Half Waist – Relaxed 1/   |          |        |         |        |        |          |      |
| X-Small                   | 12       | 12     | 12      | 12     | --     | --       | ±1/4 |
| Small                     | 14       | 14     | 14      | 14     | 14     | --       |      |
| Medium                    | 16       | 16     | 16      | 16     | 16     | 16       | ±1/2 |
| Large                     | --       | 18     | 18      | 18     | 18     | 18       |      |
| X-Large                   | --       | --     | 20      | 20     | 20     | --       |      |
| XX-Large                  | --       | --     | 22      | 22     | 22     | 22       |      |
| Half Waist – Stretched 2/ |          |        |         |        |        |          |      |
| X-Small                   | 14       | 14     | 14      | 14     | --     | --       | ±1/2 |
| Small                     | 16       | 16     | 16      | 16     | 16     | --       |      |
| Medium                    | 18       | 18     | 18      | 18     | 18     |          |      |
| Large                     | --       | 20     | 20      | 20     | 20     | 20       |      |
| X-Large                   | --       | --     | 22      | 22     | 22     | --       |      |
| XX-Large                  | --       | --     | 24      | 24     | 24     | 24       |      |
| Inseam 3/                 |          |        |         |        |        |          |      |
| X-Small                   | 28-1/2   | 30-1/2 | 32-1/2  | 34-1/2 | --     | --       | ±3/4 |
| Small                     | 28-1/2   | 30-1/2 | 32-1/2  | 34-1/2 | 36-1/2 | --       |      |
| Medium                    | 28-1/2   | 30-1/2 | 32-1/2  | 34-1/2 | 36-1/2 | 39-1/2   |      |
| Large                     | --       | 30-1/2 | 32-1/2  | 34-1/2 | 36-1/2 | 39-1/2   |      |
| X-Large                   | --       | --     | 32-1/2  | 34-1/2 | 36-1/2 | --       |      |
| XX-Large                  | --       | --     | 32-1/2  | 34-1/2 | 36-1/2 | 39-1/2   |      |
| Outseam 4/                |          |        |         |        |        |          |      |
| X-Small                   | 37-3/4   | 40-1/4 | 42-3/4  | 45-1/4 | --     | --       | ±3/4 |
| Small                     | 38-1/4   | 40-3/4 | 43-1/4  | 45-3/4 | 48-1/4 | --       |      |
| Medium                    | 39-3/4   | 41-1/4 | 43-3/4  | 46-1/4 | 48-3/4 | 52-1/4   |      |
| Large                     | --       | 41-3/4 | 44-1/4  | 46-3/4 | 49-1/4 | 52-3/4   |      |
| X-Large                   | --       | --     | 44-3/4  | 47-1/4 | 49-3/4 | --       |      |
| XX-Large                  | --       | --     | 45-1/4  | 47-3/4 | 50-1/4 | 53-3/4   |      |
| Bottom 5/                 |          |        |         |        |        |          | ±1/4 |
| X-Small                   | 8-1/2    | 8-1/2  | 8-1/2   | 8-1/2  | --     | --       |      |
| Small                     | 8-1/2    | 8-1/2  | 8-1/2   | 8-1/2  | 8-1/2  | --       |      |
| Medium                    | 8-3/8    | 8-3/8  | 8-3/8   | 8-3/8  | 8-3/8  | 8-3/8    |      |
| Large                     | --       | 8-3/8  | 8-3/8   | 8-3/8  | 8-3/8  | 8-3/8    |      |
| X-Large                   | --       | --     | 9-1/4   | 9-1/4  | 9-1/4  | --       |      |
| XX-Large                  | --       | --     | 9-1/4   | 9-1/4  | 9-1/4  | 9-1/4    |      |

1/ Half Waist – Relaxed – Measurement along center of waistband from outside folded edge to folded edge, with elastic inserts relaxed (see Figure 11).

2/ Half Waist – Stretched – Measurement along center of waistband from outside folded edge to folded edge, with half waist rotated so that elastic is in the approximate center to prevent the hands or stretching device from restricting the elastic from fully elongating. (see Figure 12).

3/ Inseam – Measurement at inseam of trousers from center of crotch seam (center of gusset) to bottom edge of trouser leg (see Figure 13).

4/ Outseam – Measurement from top edge of waist to bottom of leg along outseam.

5/ Bottom – Measurement across bottom of leg from folded edge to folded edge.



3.11 Workmanship. The finished blouse and trouser shall be free from loose thread, foreign matter, and irregular defects that can adversely affect usage or durability. The finished item shall conform to the requirements of this document, be uniform, and be free from the defects specified in **Error! Reference source not found.** and those defects that adversely affect form, fit, or function. Defects shall not exceed the Acceptance Quality Limits (AQL), as specified (see **Error! Reference source not found.**), and shall not adversely affect the serviceability, appearance, and uniformity of the product.

#### 4. VERIFICATION

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

First article inspection (see 4.2)

Quality conformance inspection (see 4.3)

4.2 First article inspection. The first article, submitted in accordance with 3.1, shall be inspected for compliance with design, configuration, workmanship, and dimensional requirements. The presence of excessive defects, as defined by contract, (see 4.1) or failure to pass any test may be cause for rejection of the first article.

4.3 Quality conformance inspection. Sampling for inspection shall be performed in accordance with ASQ/ANSI Z1.4, as defined by the contract.

4.4 Component and end item inspection. In accordance with 4.1, the components and end items shall be tested in accordance with all the requirements of referenced publications unless otherwise excluded, amended, modified, or qualified in this document or applicable procurement documents. The government reserves the right to inspect all components and end items to determine conformance to requirements. A certification of compliance may be acceptable as evidence that the permethrin treatment meets the requirements of paragraph 3.3.9. When Certificate of Compliance are submitted, the QAR will conduct in-process inspections and review records to audit compliance to the treatment process approved by the government in production verification and EPA registration. The Government reserves the right to periodically inspect such items to determine the validity of the certification in accordance with MIL-DTL-44411 and the performance requirements of this specification.

4.4.1 Breaking strength and elongation test. The thread shall meet the requirement stated in 3.4.2.1 when tested according to ASTM D2256, except testing speed shall be  $12 \pm 0.5$  in./min and a 10-inch gauge shall be used.

4.4.2 Thread colorfastness test. The thread shall meet the requirements stated in the associated thread specification.

4.4.3.3 Button pull test. The buttons attached to the blouse and trouser shall meet the requirements stated in 3.4.4 when tested according to ASTM D5034, except slide button on top of 1 inch wide grips separated at least 3/8 inch so that button loosely rests evenly on top of grips. Use manually adjusted grips only, not pneumatic. Place button attachment stitching in middle of bottom grips and adjust an additional 3/8-inch down so that the grips grab cloth just below stitching, not on the stitching. Run tensile machine per test method and record force required to pull button from sewn garment.

4.4.4 Toxicity test. The finished blouse and trouser shall be tested for dermal toxicity as follows:

Title 40 , Code of Federal Regulations, 1994 Edition; Part 798.4100 – Dermal Sensitization

Part 798.4470 – Primary Dermal Irritation Part 798.4500 – Primary Eye Irritation

As an alternative to testing, the contractor may provide information, which certifies that the treated cloth is composed of chemicals and/or materials, which have been safely used commercially where prolonged, repeated skin contact has occurred.

4.4.5 Material and garment testing. The cloth and/or garment shall be tested for characteristics listed in 3.3. The testing shall be performed using the test methods as specified in table XVI. All test reports should contain the individual values utilized in expressing the final results. For material testing, the sample unit shall be five continuous yards full width of the finished cloth, for all physical and chemical tests. For garment testing, the sample unit shall be one blouse and one trouser. The lot shall be considered unacceptable If one or more sample units fail to meet any requirements specified.

TABLE XVI. Material and Garment Testing Requirements.

| CHARACTERISTIC                                                                                         | REQUIREMENT<br>PARAGRAPH | TEST METHOD                     |
|--------------------------------------------------------------------------------------------------------|--------------------------|---------------------------------|
| Garment Testing:                                                                                       |                          |                                 |
| pH                                                                                                     | 3.4                      | AATCC 81                        |
| Toxicity                                                                                               | 3.3.3                    | 4.4.4 <u>1/</u>                 |
| Elastic Force vs Elongation                                                                            | 3.6.3                    | 4.4.5.4                         |
| Cargo Pocket                                                                                           |                          | 4.4.5.4.1                       |
| Waistband                                                                                              |                          | 4.4.5.4.2                       |
| Seam Strength (min)                                                                                    | 3.4.2                    | ASTM D 5034 <u>1/</u>           |
| Permethrin Content and bite protection                                                                 | 3.3.11                   | 4.4.5.5                         |
| Flammability (after 1 and 25 laundering cycles)<br>Instrumented Manikin Test (4 second flame exposure) | 3.3.1 and 3.4.1          | ASTM F1930<br>4.4.5.8 <u>2/</u> |
| Material Testing:                                                                                      |                          |                                 |
| Dimensional stability (max)                                                                            | 3.3.10                   | AATCC 96                        |
| Weight, ounces per square yard                                                                         | 3.3.1                    | ASTM D3776/D3776M, Option C     |
| Fiber content                                                                                          | 3.3.1                    | AATCC 20A <u>1/</u>             |
| Weave                                                                                                  | 3.3.1                    | Visual <u>1/</u>                |
| Breaking Strength, pounds (min)                                                                        | 3.3.1                    | ASTM D 5034                     |
| Tear Strength, pounds (min)                                                                            | 3.3.1                    | ASTM D1424                      |
| Knit construction                                                                                      | 3.3.1                    | Visual <u>1/</u>                |
| Antimicrobial properties<br>Initial (no laundering) and after 25 laundering cycles                     | 3.4.3                    | AATCC 100 <u>1/</u>             |
| Bursting Strength, pounds (min)                                                                        | 3.3.1                    | ASTM D 6797                     |
| Air Permeability, cfm (min)                                                                            | 3.3.1                    | ASTM D 737                      |
| Drying Time, minutes (max)                                                                             | 3.3.1                    | Internal Method                 |

TABLE XVI. Material and Garment Testing Requirements. Continued

| Characteristic                                                                   | Requirement Paragraph | Test Method                                                   |
|----------------------------------------------------------------------------------|-----------------------|---------------------------------------------------------------|
| Colorfastness                                                                    | 3.3.1                 | AATCC 61 Test 1A<br>AATCC 16.2 or 16.3<br>AATCC 15<br>AATCC 8 |
| Laundering (after 4 cycles)                                                      |                       |                                                               |
| Light (40 hours or 17 kilojoules)                                                |                       |                                                               |
| Perspiration (acid and alkaline)                                                 |                       |                                                               |
| Crocking                                                                         |                       |                                                               |
| Visual Color Matching                                                            | 3.3.6.1               | AATCC Evaluation Procedure 9, Option A, 4.4.5.2.1             |
| Instrumental Color Matching                                                      | 3.3.6.2               | AATCC Evaluation Procedure 6, 4.4.5.2.2                       |
| Moisture Vapor Transmission, per square meter per 24 hours (min)                 | 3.3.1                 | ASTM E 96 Test B                                              |
| Pilling                                                                          | 3.3.1                 | ASTM D3511,<br>AATCC 96                                       |
| Spectral Reflectance                                                             | 3.3.9                 | 4.4.5.1                                                       |
| Thermal Protective Performance, cal/cm <sup>2</sup> (min)                        | 3.3.1                 | NFPA 1971 1/                                                  |
| Thermal Shrinkage (warp and fill)                                                | 3.3.1                 | NFPA 1971 1/                                                  |
| Vertical Flame<br>After Flame, seconds (max) Char Length, inches (max) Melt/Drip | 3.3.1                 | ASTM D6413                                                    |
| Presence of labile sulfur                                                        | 3.3.3                 | 1/                                                            |

1/ When specified (see 6.2), Certificates of Compliance (CoC) may be submitted and will be acceptable for the stated requirement. When CoC are submitted the Government reserves the right to inspect such items to determine the validity of the certification. If a significant change is made to the garment design or materials (as determined by the Marine Corps Systems Command (MARCORSSYSCOM)) verification testing may be conducted.

2/ The Government has verified that garments constructed with specified fabrics meet the Instrumented Manikin Flammability requirements. If any changes are made to the specified fabrics (knit or woven) or design, the Government reserves the right to conduct verification testing of the flammability requirements

4.4.5.1 Spectral Reflectance. Initial and permethrin treated cloth shall meet requirements specified in 3.3.7. Spectral reflectance shall be measured and reported on the initial cloth. Certificate of compliance will be accepted on finished uniforms and subject to Government verification. Spectral reflectance data shall be determined on the face side and shall be obtained from 600 to 860 nanometers (nm) at 20 nm intervals on a spectrophotometer relative to the barium sulfate standard, the preferred white standard. Other white reference materials may be used provided they are calibrated to absolute white, e.g. magnesium oxide or vitrolite tiles. The spectral bandwidth shall be less than 26 nm at 860 nm. Reflectance measurements may be made by either the monochromatic or polychromatic mode of operation. When the polychromatic mode is used, the spectrophotometer shall operate with the specimen diffusely illuminated with the full emission of a source that simulates either CIE source A or CIE source

D65. The specimen shall be measured as a single layer, backed with six layers of the same cloth and shade. Measurements shall be taken on a minimum of two different areas and the data averaged. The measured areas should be taken at least 6 inches away from the selvage. The specimen shall be viewed at an angle no greater than 10 degrees from the normal, with the specular component included. Photometric accuracy of the spectrophotometer shall be within 1 percent and the wavelength accuracy within 2 nm. The standard aperture size used in the color measurement device shall utilize the largest aperture permitted by relevant color patch on a selected cloth (either 0.25 or 0.50 inches in diameter). Any color having spectral reflectance values outside the limits at four or more of the wavelengths specified shall be considered a test failure.

#### 4.4.5.2 Color Matching.

4.4.5.2.1 Visual color matching (all cloth). The color and appearance of all cloth, camouflage printed and permethrin treated cloth and dyed knit cloth, shall match the standard sample when viewed using AATCC Evaluation Procedure 9, Option C; under filtered tungsten lamps that approximate artificial daylight D65 illuminant with a color temperature of  $6500 \pm 200$  Kelvin with illumination of  $100 \pm 20$  foot candles, and shall be a good match to the standard sample under horizon lamplight at  $2300 \pm 200$  Kelvin.

4.4.5.2.2 Instrumental color matching (all cloth). Instrumental color matching is used as a tool to quantify shade evaluation if visually shade is rated unacceptable. All the colors in the permethrin treated Woodland MARPAT shall be instrumentally measured except for Khaki and all the colors in the permethrin treated Desert MARPAT shall be measured except for Highland given the areas of these exempted colors are too small for accurate instrumental readings. The solid colored permethrin treated knit cloth shall be instrumentally measured. Each measured color shall match the standard sample using AATCC Evaluation Procedure 6. A color difference greater than a  $\Delta E$  CMC = 1.5, when using a  $\Delta E$  CMC (2:1) ratio (D65 /  $10^\circ$ ) units as compared to the standard sample, shall be basis for rejection.

4.4.5.3 Toxicity assessment. The contractor must furnish information (see 4.4.5.3.1) certifying that the finished product is composed of materials which have been safely used commercially OR which provide sufficient toxicity data to show compatibility with prolonged, direct skin contact. At a minimum, toxicity data should include results from a primary dermal irritation study in laboratory animals and a repeated insult human patch test (Modified Draize Procedure). The latter must be conducted under the supervision of a qualified dermatologist using at least 100 free-living individuals.

4.4.5.3.1 Toxicity Documents. All finishes/chemicals used to process the garment shall be identified and accompanied by the appropriate Material Safety Data Sheet (MSDS) information. The use of chemicals recognized by the Environmental Protection Agency (EPA) as known human carcinogens is prohibited.

4.4.5.4 Force versus elongation of elasticized features. Cargo pocket and waistband samples shall be samples from garments that are constructed to duplicate the configuration on the finished garment where the elastic webbing is set on or within the trouser cloth. The cargo pocket length (relaxed) is the finished opening length of six inches in length. The waistband shall consist of one side of the elasticized waistband which is approximately four inches in length (relaxed). Both cargo pocket and waistband samples are laundered 5 times as per AATCC 135, 3, V, iii and stitching which sets the elastic must stay secured through the washings. Tensile properties are evaluated by conducting tensile tests utilizing equipment, which conform to ASTM D76, Standard Specification of Tensile Testing Machines for Textiles. Samples are conditioned in a 70 degree F, 65percent RH laboratory for a minimum of 24 hours. The specimens are clamped in pneumatic jaws in a relaxed state (no force on read out) where the jaws are spaced 4.0 inches apart.

4.4.5.4.1 Cargo Pocket. The cargo pocket sample shall be centered between the jaws so that the set elastic webbing is clamped into each jaw. A Constant Rate of Extension (CRE) with a loading rate of 20.0 inches per minute is used. The crosshead is moved 1.5 inches for a total of 5.5 inches. The resulting force is recorded at a rate of 10 points per second. One cycle consists of this elongation as well as the

return to the initial 4 inch separation position. The cycle is repeated 100 times for each sample. The graph shall be evaluated and the sample will meet the requirement when no peak is greater than 3.5 lbs. nor less than 2.5 lbs.

4.4.5.4.2 Waistband. A Constant Rate of Extension (CRE) with a loading rate of 2.0 inches per minute is used. The crosshead is moved 1.5 inches for a total of 5.5 inches. The resulting force is recorded at a rate of 10 points per second. The graph shall be evaluated to record the force separately at 3/4 inch and 1-1/2 inch extension. The sample meets requirements when the 3/4 inch extension does not exceed 1.5 pounds nor is less than 0.5 lbs, and the 1-1/2 inch extension does not exceed 4 pounds nor less than 1-1/2 pounds. The sample must demonstrate acceptable performance at both extensions to pass the requirement.

4.4.5.5 Insect repellent treatment verification. Verification of the insect repellent treatment shall be in accordance with MIL-PRF-32659. The contractor shall notify and request Government approval any time there is a change in the treatment formulation or processing conditions. Changes shall include manufacturing supply chain, active ingredients, and application method. Insect repellent active ingredient treatment level and percent mosquito bite protection shall be verified before the Government approves the change.

4.4.5.6 Instrumented Manikin Testing. Finished end item blouse and trousers shall be tested according to ASTM F1930 and subjected to 4 second exposure at 2.0 cal/cm<sup>2</sup>/sec heat flux. The EFRCE (medium-regular size) shall be tested after one and twenty-five laundering cycles according to AATCC 135, 3, V, Aiii. Items shall be tested with cotton t-shirt and briefs underneath the EFRCE. The EFRCE shall have no more than 25 percent total burn injury prediction including second- and third-degree burns. The burn sensors in the torso and head sections should be excluded from the total burn injury calculation as the torso sensors would not register a burn as this area is covered and protected by a ballistic vest and the uniform does not cover the head. This test shall be conducted during First Article Testing (FAT) and when a significant change is made to the garment design or materials. The change is determined to be significant by the Marine Corps Systems Command.

4.4.5.9 Microbe growth (knit cloth only). Microbe growth shall be measured in accordance with AATCC TM100 and strict adherence to the additional requirements specified herein to achieve correlation with Government verification tests. No substitutions are allowed.

4.4.6 End item visual examination. Finished end item blouse and trousers shall be subjected to visual examination after permethrin treatment. All fabric and garment defects shall be scored in accordance with examination descriptions as specified in table XIII.

TABLE XIII. End Item Visual Examination.

| Examination              | Defect Description                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Bartacks                 | Bartacks or bartacks missing, insecure, misplaced, not specified size, stitches loose or broken, bartack/backtack not serving intended purpose                                                                                                                                                                                                                                                                                                                                   |
| Belt Loops               | Belt loops omitted, insecure, not specified size, or opening out of tolerance.                                                                                                                                                                                                                                                                                                                                                                                                   |
| Buttons &<br>Buttonholes | Trouser fly and pocket flap buttons and button holes out of alignment causing bulge, twist or distortion when buttoned.                                                                                                                                                                                                                                                                                                                                                          |
|                          | Buttonholes and eyelets omitted, added, not clean cut or securely caught in fabric, not specified type, not specified location, not positioned as specified, buttonhole purling not positioned as specified, button not properly attached.                                                                                                                                                                                                                                       |
|                          | Sleeve buttonhole placement not as specified – first buttonhole not aligned with cuff tab button, second buttonhole less than 3 1/8 or more than 3-3/8 inches from first, third button hole less than 1-7/8 or more than 2-1/8 inches from second. Eyelet end of sleeve buttonholes less than 5/8 inch or more than 7/8 inch from bottom edge of sleeve hem. Cuff button hole opening greater than 7/8 inch. Buttonhole stitching reversed purling not stitched on correct side. |
|                          | Button color not as specified                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Component Part           | Component part of blouse or trouser omitted, distorted, full, tight, or twisted; any part of louse or trouser caught in any unrelated stitching, the edge of any component part required to be forced out having folds of more than 1/8 inch. Fullness creating unwanted permanent fold, pleat, or crease in fabric or garments, shade variations within or between parts. <u>1/</u>                                                                                             |
| Elastic                  | Trouser waist elastic omitted, not positioned as specified, not attached as specified, width not as specified, not caught in bartack, bartack not required length or positioned as specified                                                                                                                                                                                                                                                                                     |
|                          | Trouser cargo pocket elastic does not expand/retract to relaxed condition as specified, omitted, twisted, width not as specified, not caught in bartack, bartack not length or position specified, setting distorts trouser leg.                                                                                                                                                                                                                                                 |
| Evenness <u>2/</u>       | Blouse:                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|                          | Blouse collar front points vary by more than 1/8 inch, collar curls, puckers, pleats, or twists. End of collar and edge of front facing out of alignment by more than 1/8 inch.                                                                                                                                                                                                                                                                                                  |
|                          | Sleeve lengths vary by more than 1/2 inch. Cuff and cuff tab out of alignment with bottom folded edge of sleeve hem by more than 1/8 inch.                                                                                                                                                                                                                                                                                                                                       |
|                          | Trouser:                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                          | Waistband uneven more than 1/4 inch when buttoned                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                          | Inseam and/or outseam lengths vary by more than 1/2 inch from leg to leg                                                                                                                                                                                                                                                                                                                                                                                                         |
|                          | Bottom openings vary by more than 1/2 inch in half width                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                          | Evenness of length between inseam and outseam varies by more than 1/2 inch                                                                                                                                                                                                                                                                                                                                                                                                       |

TABLE XIII. End Item Visual Examination. Continued

| Examination       | Defect Description                                                                                                                                                                                                                                               |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Eyelets           | Omitted, misplaced, improper size or caught in stitching                                                                                                                                                                                                         |
| Hems              | Hems at blouse bottom, sleeves, and/or trouser legs twisted, wavy, omitted or not as specified.                                                                                                                                                                  |
|                   | Hem width of blouse bottom and trousers less than 1/2 inch or more than 3/4 inch. Sleeve hem width less than 2-3/8 inch or more than 2-5/8 inch. Hem measurement taken from top fold to bottom fold.                                                             |
| Loop              | Loop color or size not as specified                                                                                                                                                                                                                              |
| Labels            | Omitted, incorrect, illegible, not attached where specified; bar-codes omitted, not readable by scanner; human-readable interpretation (HRI) omitted or illegible; bar code not visible on folded, packaged item; bar code attachment causes damage to the item. |
| Material          | Hole, slub, cut, tear, smash, burn, exposed drill hole, run, thin place, color/dye streak, spots and/or stains, slubs, knots, color not as specified, misweave <u>3</u> /                                                                                        |
|                   | Fabric used for each blouse and trouser not as specified.                                                                                                                                                                                                        |
| Patches           | Elbow, seat and knee patches omitted, not attached as specified, or not positioned as specified in pattern                                                                                                                                                       |
| Packaging         | Any blouse or trouser not packaged in accordance with the contract or purchase order                                                                                                                                                                             |
| Pleats            | Trouser pleats omitted, reversed, i.e. not facing outseam, front panel pleats stitched down less than 2-3/4 or more that 3-1/4 inches from top of trousers.                                                                                                      |
| Pockets and Flaps | Pocket companions not uniform in size or shape                                                                                                                                                                                                                   |
|                   | Pockets twisted, curled or puckered, not stitched or located as specified                                                                                                                                                                                        |
|                   | Pocket flap not completely covering pocket opening, not positioned as specified.                                                                                                                                                                                 |
|                   | Pocket flaps not centered over pocket opening. Flap width extends pass finished pocket edge by more than 1/8 inch on each side.                                                                                                                                  |
|                   | Sleeve and cargo pocket bellows exposed beyond edge of pockets by more than 1/8 inch                                                                                                                                                                             |
|                   | Sleeve, cargo, and hip pocket out of horizontal alignment by more than 1/2 inch                                                                                                                                                                                  |
|                   | Sleeve pocket flap length at center from setting seam greater than 2-3/4 inches                                                                                                                                                                                  |
|                   | Large Cargo pocket elasticized opening less than 6-3/4 or more than 7-1/2 inches in width, forward bottom corner of flap not bartacked down. Bartack not positioned and stitched through both flap and pocket.                                                   |
|                   | Side Hanging pocket opening less than 6 inches.                                                                                                                                                                                                                  |

TABLE XIII. End Item Visual Examination. Continued

| Examination      | Defect Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Pressing Defects | Unwanted permanent fold, pleat or crease in garments affecting appearance or serviceability.                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Stitching        | Blouse or trouser seam: open stitching, puckered, distorted, pleated, repaired, wavy, twisted, irregular, loose or tight stitch tension, broken or missing thread or stitch, fullness, needle chew, visible mend, edge or raised stitching sewn too close to edge resulting in damage to cloth, seam allowance not as specified, visible raw edge on outside of garment, raw edge on outside, raw edge on inside along double needle seams, raw edge greater than 1/8 inch on inside of garment (to include pockets). |
|                  | Stitching not as specified.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                  | Double needle intersecting seams staggered by more than 1/4 inch except the crotch seam which is staggered by more than 1/2 inch.                                                                                                                                                                                                                                                                                                                                                                                     |
|                  | Run off of more than 1/2 inch for edge and raised stitching                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                  | Thread color not as specified                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

1/ Parts suspected of being off shade shall be examined at a distance of 3 feet against a background of the other parts and colors of the garment. When the shade difference is discernible under these examination conditions, it shall be scored as a shaded part.

2/ See 4.4.7.1.

3/ As defined in ASTM D3990.

4.4.7 Finished garment visual examination. The garment shall be inspected after it is gently smoothed by hand and lying flat. Defects are defined in paragraph 4.4.6.

4.4.7.1 Evenness. Evenness can be determined by comparing measurements of companion part OR by aligning parts and measuring difference in lengths as described below. Except for the collar, comparisons shall be inspected by extending the portions of the garments in parallel between two hands without stretching and inspecting for evenness. Where evenness defects are found, the garments shall be laid flat on a measuring surface and the difference in lengths recorded.

Collar: Evenness shall be assessed by folding the collar at center back and align the collar halves at the setting seam to the front collar ends. Rounded front collar points shall be compared for height, shape and curve.

Sleeves: Sleeves shall be aligned from top of shoulder seam, smoothed down along sleeve crease to sleeve hem edge. The aligned sleeve lengths shall be visually inspected for evenness of sleeve lengths, and sleeve pocket placement.

Inseam: Inseam shall be aligned with inseam along the top edge, starting from the center of the crotch seam along the inseam down to the trouser hem edge. The aligned trouser legs shall be inspected for evenness of the trouser legs.

Outseam: Outseam alignment shall be aligned by folding trousers in half width wise following the permanent press crease of the legs and aligning the top of the waistband. The aligned trouser legs shall be inspected for evenness of trouser length, and placement of companion cargo pockets. The



folded trouser shall be turned vertically to enable alignment of the bottom opening widths for comparison.

Inseam/Outseam Length: While measuring inseam, the uniformity in length between inseam and outseam shall be inspected for evenness.

## 5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or purchase order (see 6.2). When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

5.2 Permethrin packaging. Every box containing permethrin treated uniforms must be labeled according to EPA requirements as stated in the Federal Insecticide, Fungicide, and Rodenticide Act (FIFRA) (see paragraph 2.2.2).

## 6. NOTES

(This section contains information of general or explanatory nature that may be helpful, but is not mandatory.)

### 6.1 Intended use.

### 6.2 Acquisition requirements. Acquisition documents should specify the following:

- a. Title, number and date of this document.
- b. Types, classes and sizes required (see 1.2).
- c. Revision or publication date of publications cited in the solicitation and, if required, the specific issue of individual publications referenced (see 2.2. and 2.3).
- d. When first article sample is required (see 3.1, 4.2 and 6.3).
- e. Quality conformance inspection requirements (see 3.1 and 4.3).
- f. When recycled materials are prohibited (see 3.2).
- g. Any approved deviations from the specified physical requirements of the materials, components or design.
- h. Acceptance quality limits (AQL) and when certificates of compliance will be accepted
- i. Packaging requirements (see 5.1).
- j. Inspection level (see 4.1 and 4.3)

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209-4. The first article should be a pre-production sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all acquisitions documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Suggested Sources.6.4.1 Woven.

Milliken & Company  
 920 Milliken Road, M-159 Spartanburg, SC 29304  
 (864)503-1765  
 Milliken style ResQ II 6.7 oz/sq yd or equal.

6.4.2 Knit.

SSM Industries, Inc.  
 211 Ellis Avenue  
 Spring City, TN 37381  
 (423)365-7181  
 (Sigma 4 Star Knit Style SD-1883.00)

6.4.3 FR foam.

UFP Technologies Fortiform

6.5 Figures. See attached Figures 1-13.

6.6 Size abbreviation. The size abbreviation on the garment label shall show the combination of one of each of the following size and length abbreviations shown below. The combination of the two elements shall be designated as Size -(hyphen) Length; for example Small Regular abbreviated designation is Sm-Reg, X-Small X-Short abbreviated designation is XSm-XSht, etc.

| Size     | Size Abbreviation | Length   | Length Abbreviation |
|----------|-------------------|----------|---------------------|
| X-Small  | as XSm            | XX-Short | as XXSht            |
| Small    | as Sm             | X-Short  | as XSht             |
| Medium   | as Med            | Short    | as Sht              |
| Large    | as Lg             | Regular  | as Reg              |
| X-Large  | as XLg            | Long     | as Lng              |
| XX-Large | as XXLg           | X-Long   | as XLng             |
|          |                   | XX-Long  | as XXLng            |

6.7 Blouse size/identification/care label example.

Medium - X-Short  
 Height: 59 to 63 in.  
 Chest: 37 to 41 in.  
 Stock No: 8415-01-484-5933  
 Enhanced FR Combat Ensemble Blouse, Woodland MARPAT™  
 Fiber Content  
 U. S. Patent Nos. D464.790S & D491.372S. CONTRACT #: M#####-##-X-####  
 Apparel USA

Care Information:

DO NOT BLEACH, STARCH, DRY CLEAN, OR PRESS

1. Washing. Machine wash using Permanent Press Cycle or hand wash in warm water using mild detergent that does NOT contain optical brighteners. Rinse completely. Do not overload the machine.
2. Drying. Tumble dry on low heat. Do not overload the dryer.
3. Fabric softener. The use of fabric softeners is not recommended due to potential to adversely affect the flame protection.

6.8 Trouser size/identification/care label example:

Medium - Short

Waist: 27 to 31 in..

Inseam: 26-1/2 to 29-1/2 in. Stock No: 8415 01 484 5933

Enhanced FR Combat Ensemble Trousers, Woodland MARPAT™ Fiber Content

U. S. Patent Nos. D464.790S & D491.372S. CONTRACT #: DAAD16-01-D-9999

Apparel USA

6.9 Insect protection label example. The garments shall include a permanent insect protection label complying with the approved EPA registration. Example of a label approved by EPA is included in EPA web site at: <http://oaspub.epa.gov/pestlabl/ppls.home> and as shown below. The garment label shall include the following information:

PRODUCT BRAND NAME INSECT REPELLENT APPAREL

Refer to paper hang tag for more information.

BLOUSE or TROUSER, ENHANCED FLAME RESISTANT COMBAT ENSEMBLE

WOODLAND or DESERT CAMOUFLAGE

PERMETHRIN CONTRACT No. : COMPANY NAME (APPLIER): PRODUCT NAME:

EPA REG. NO. : EPA EST. NO. :

---

Do Not Dry Clean, Dry Cleaning removes active ingredient.

Wash separately from other clothing.

Do Not Re-treat with a permethrin product.

Dispose of garment in trash.

Repels mosquitoes. Repellency remains effective for 25 washings

ACTIVE INGREDIENT:       percent W/W Permethrin.....0.52 percent

OTHER INGREDIENTS: (GARMENT).....99.48 percent

TOTAL.....100.00 percent

It is a violation of Federal Law to use this product in a manner inconsistent with its labeling.

Retain paper hangtag for future reference on proper handling of this garment.

6.10 Approved EPA permethrin registrations. Permethrin treatment operation for the subject uniforms shall be EPA registered (Such as but not limited to EPA Registration No. 07843-0002 and No. 082392-0001).

NOTE: EPA registration does not certify that the permethrin treatment meets the EFRCE specification requirements.

6.11 GOVERNMENT INTELLECTUAL PROPERTY AND TRADEMARK RIGHTS NOTIFICATION:

This notice is to advise vendors that the Government possesses intellectual property/trademark rights in the following Marine Corps patterns and logos (hereafter collectively referred to as “intellectual property”): The Eagle, Globe and Anchor (EGA) logo, including the EGA logo as it appears embedded in the fabric pattern. The Government further has title to the invention disclosed and claimed in United States Design Patent No. D491,372 issued on 15 June 2004 for “Camouflage Pattern for Sheet Material and Uniforms” and the invention disclosed and claimed in United States Patent No. 6,805,957 issued on 19 October 2004 for ‘Camouflage U.S. Marine Corps Utility Uniform: Pattern, Fabric, and Design.’ The Government claims exclusive ownership of the above mentioned intellectual property. Therefore, no entity other than the Government, or those contracted by or having obtained proper permission or licenses from the Government to do so, are permitted to produce, sell, or transfer in any manner any items (clothing or non-clothing) containing or copying, in whole or in part, the intellectual property. Doing so will be considered an infringement on the Government’s intellectual property rights and will be subject to legal action.

6.13 Key terms.

Insect repellent  
MARPAT  
Meta-Aramid  
NWU  
Para-Aramid  
Permethrin

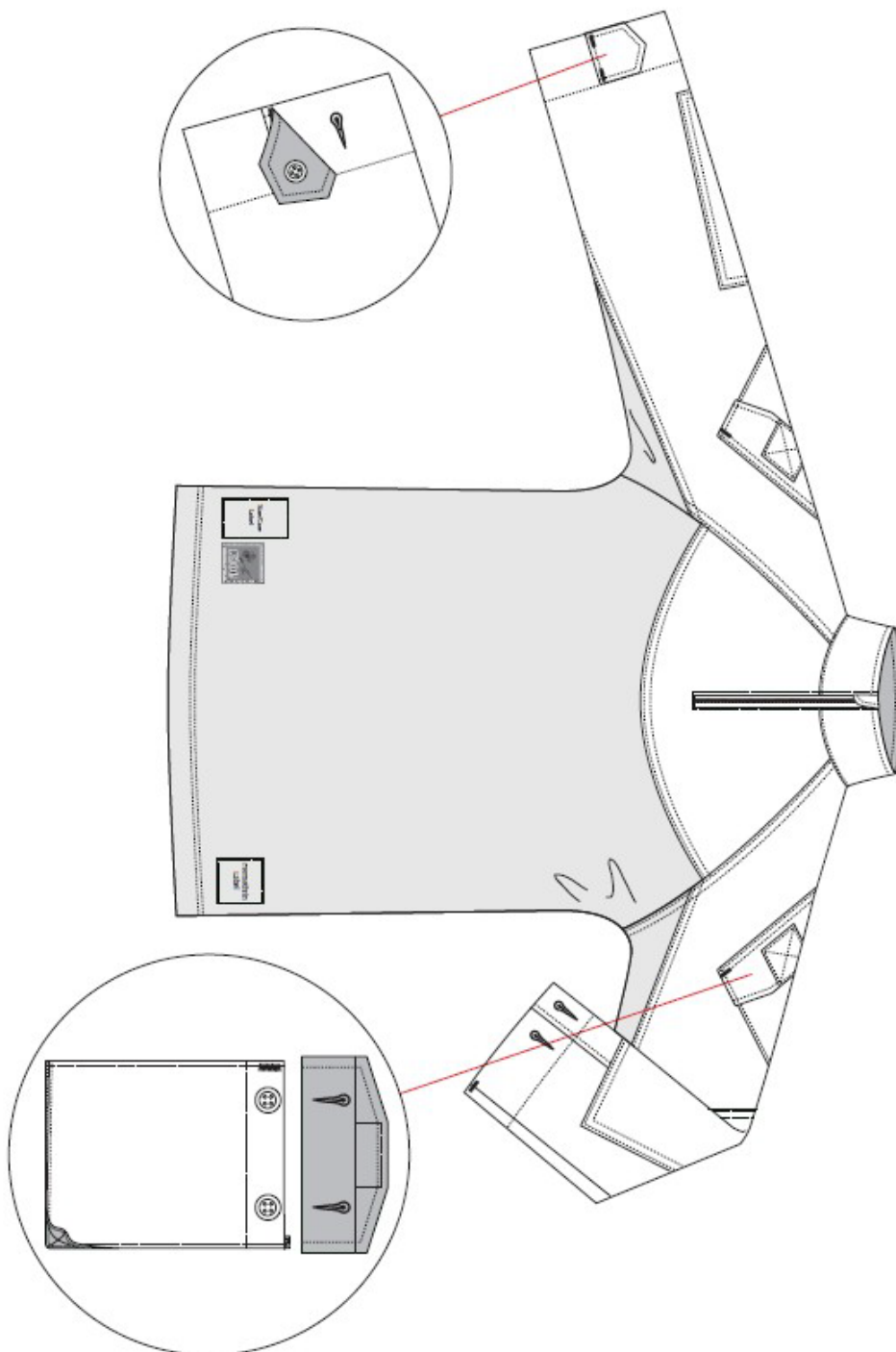


FIGURE 1. Blouse, Front View with pocket and cuff detail.

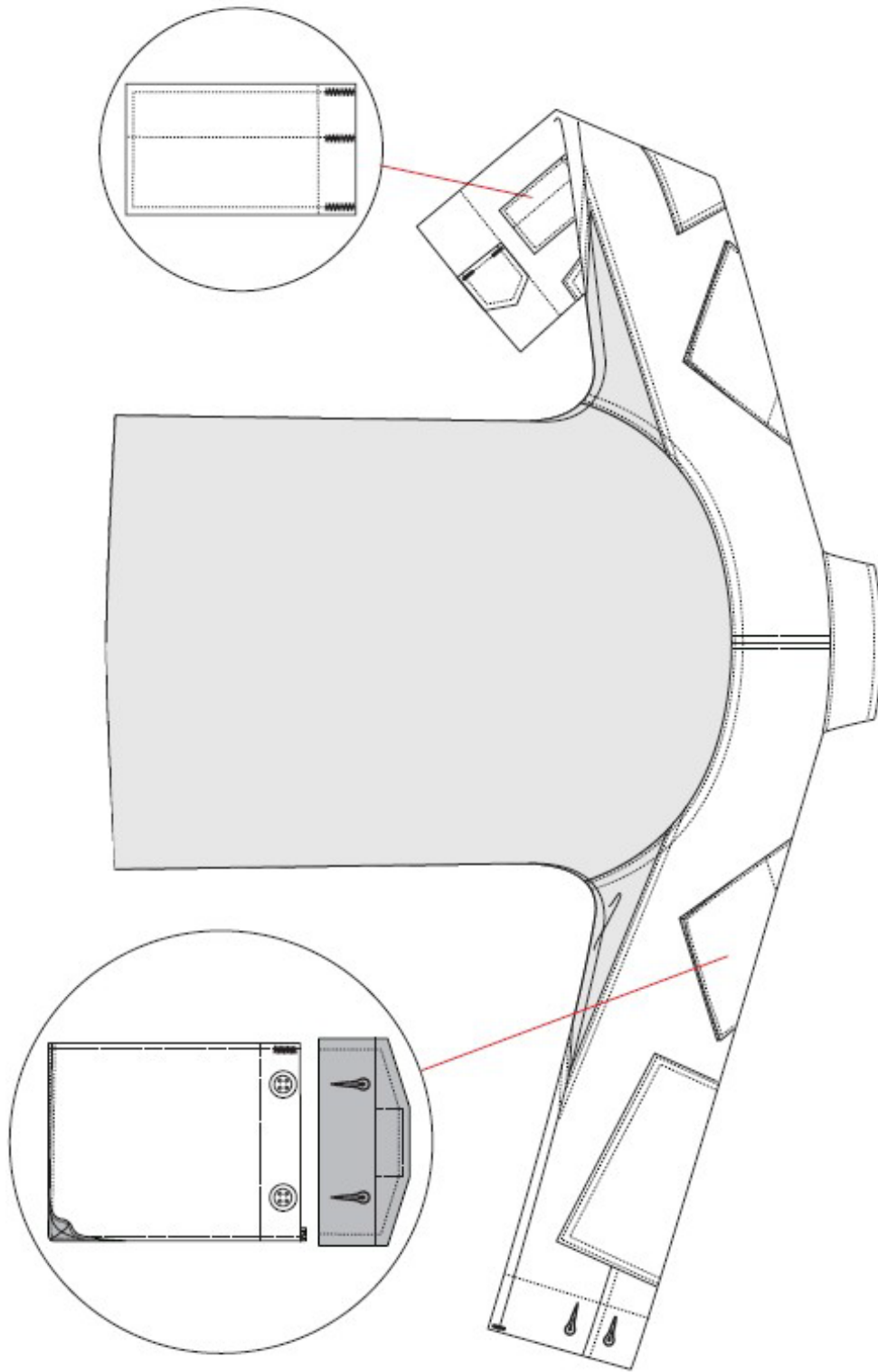


FIGURE 2. Blouse, Rear View with pen and sleeve pocket detail.

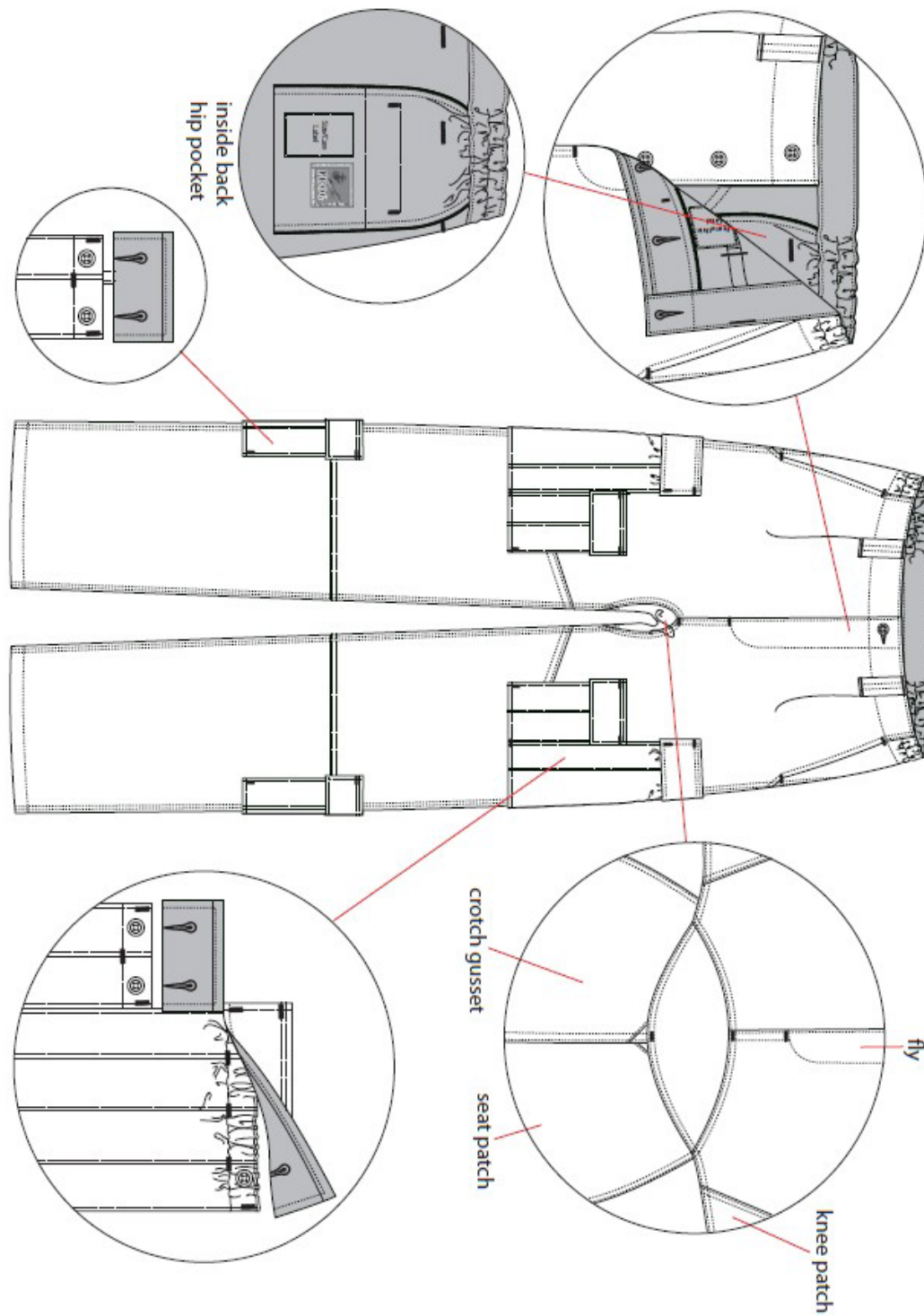


FIGURE 3. Trousers, Front View with pocket, crotch gusset, and label details.

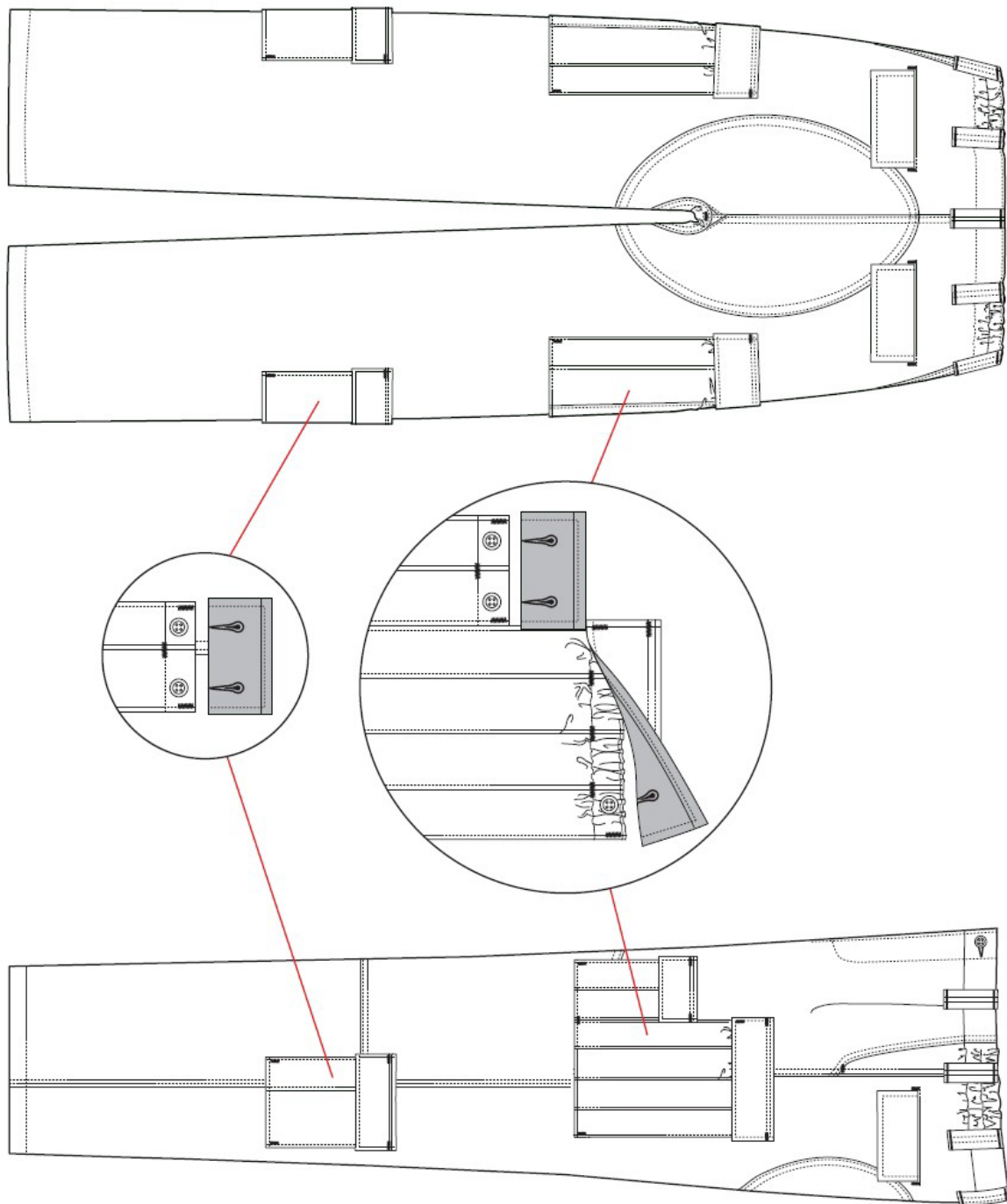


FIGURE 4. Trousers, Back and Side view with pocket detail.



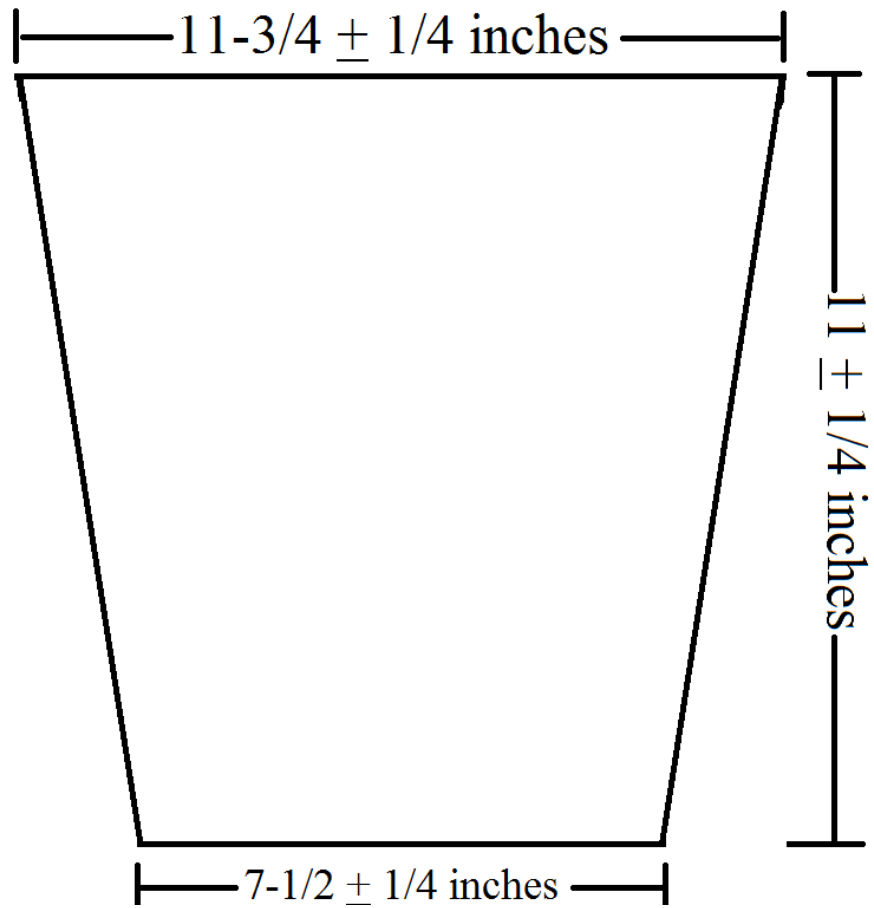


FIGURE 5. Test Specimen, percent Bite Protection Test.

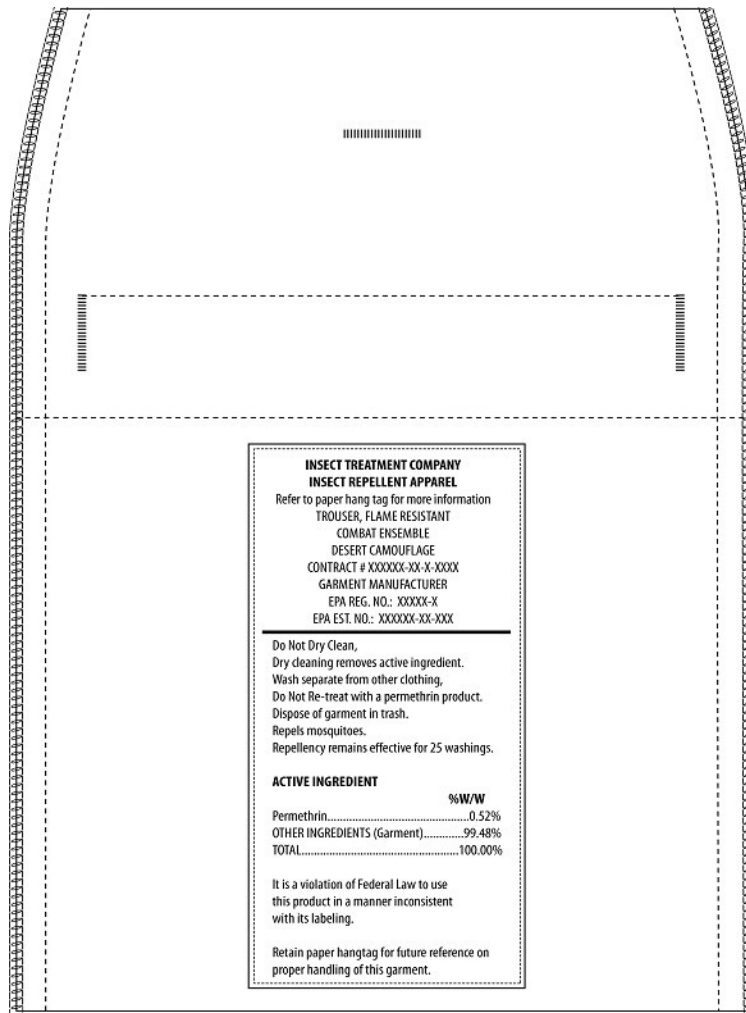
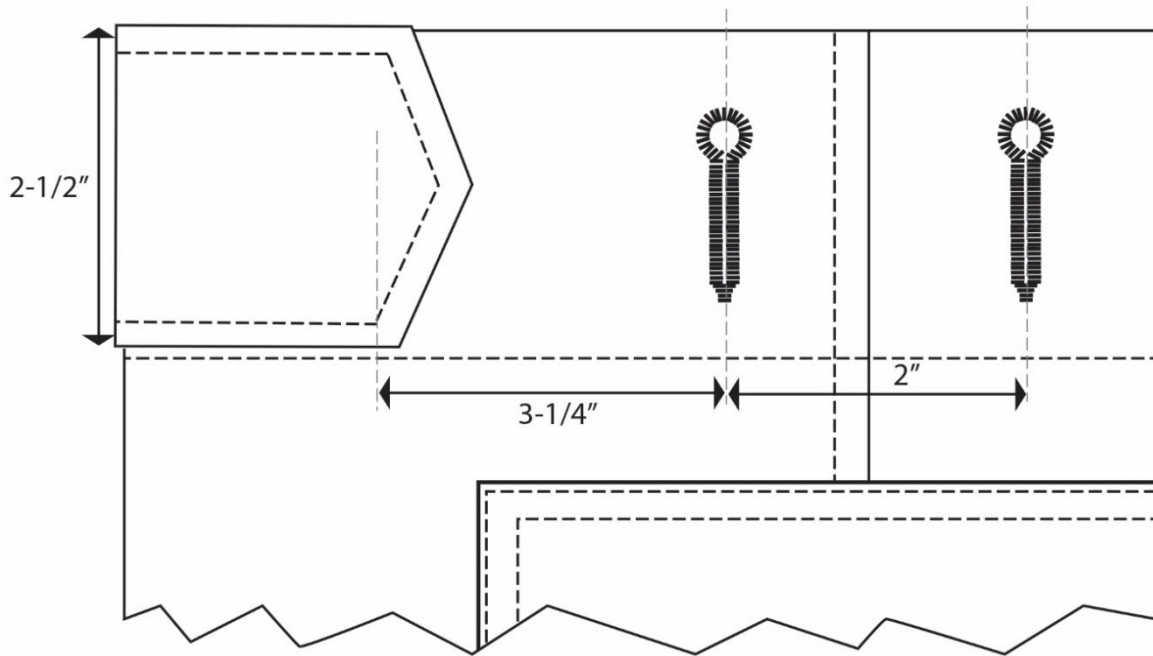


Figure 6. Insect Protection/Brand Label Placement – Right Rear Pocket Bag.

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                          |             |                 |       |                                  |        |            |         |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|-------------|-----------------|-------|----------------------------------|--------|------------|---------|
| <p style="text-align: center;">○</p> <p style="text-align: center;"><b>INSECT TREATMENT COMPANY</b><br/> <b>INSECT REPELLENT APPAREL</b><br/> TROUSER, FLAME RESISTANT COMBAT ENSEMBLE<br/> DESERT CAMOUFLAGE<br/> GARMENT MANUFACTURER<br/> INSECT TREATMENT COMPANY<br/> EPA REG. NO.: XXXXX-X<br/> EPA EST. NO.: XXXXX-XX-XXX</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                          |             |                 |       |                                  |        |            |         |
| <p>*Do Not Dry Clean,<br/> Dry cleaning removes active ingredient.<br/> *Wash separately from other clothing.<br/> *Only launder permethrin treated and untreated garments together<br/> under field or combat conditions.<br/> *Do Not Re-treat with other permethrin products.<br/> *Dispose of garment in trash. Do not contaminate water, food or feed by<br/> storage or disposal. Do not reuse clothing/fabric for purposes other<br/> than originally intended.</p> <p>Repels mosquitoes.<br/> Repellency remains effective for 25 washings</p> <table> <tr> <td><b>ACTIVE INGREDIENT</b></td> <td><b>%W/W</b></td> </tr> <tr> <td>Permethrin.....</td> <td>0.52%</td> </tr> <tr> <td>OTHER INGREDIENTS (Garment).....</td> <td>99.48%</td> </tr> <tr> <td>TOTAL.....</td> <td>100.00%</td> </tr> </table> <p>It is a violation of Federal Law to use this product in a manner<br/> inconsistent with its labeling.</p> <p>Retain hangtag for future reference on proper handling of this garment.</p> | <b>ACTIVE INGREDIENT</b> | <b>%W/W</b> | Permethrin..... | 0.52% | OTHER INGREDIENTS (Garment)..... | 99.48% | TOTAL..... | 100.00% |
| <b>ACTIVE INGREDIENT</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>%W/W</b>              |             |                 |       |                                  |        |            |         |
| Permethrin.....                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 0.52%                    |             |                 |       |                                  |        |            |         |
| OTHER INGREDIENTS (Garment).....                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 99.48%                   |             |                 |       |                                  |        |            |         |
| TOTAL.....                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 100.00%                  |             |                 |       |                                  |        |            |         |

Figure 7. Garment Hang Tag Example.



1. One buttonhole is aligned with button positioned on underside of sleeve tab.  
(Note: Buttonhole is hidden under tab in this figure)
2. Second Buttonhole is positioned  $3\frac{1}{4} \pm \frac{1}{8}$  inches from the center of first buttonhole.
3. Third Buttonhole shall be  $2 \pm \frac{1}{8}$  inches from the center of the second buttonhole.
4. All Buttonholes are positioned with cut opening  $1\frac{1}{8} \pm \frac{1}{8}$  inches from bottom edge of sleeve.

Figure 8. Wrist Closure.



Figure 9. Half Chest Finished Dimension.

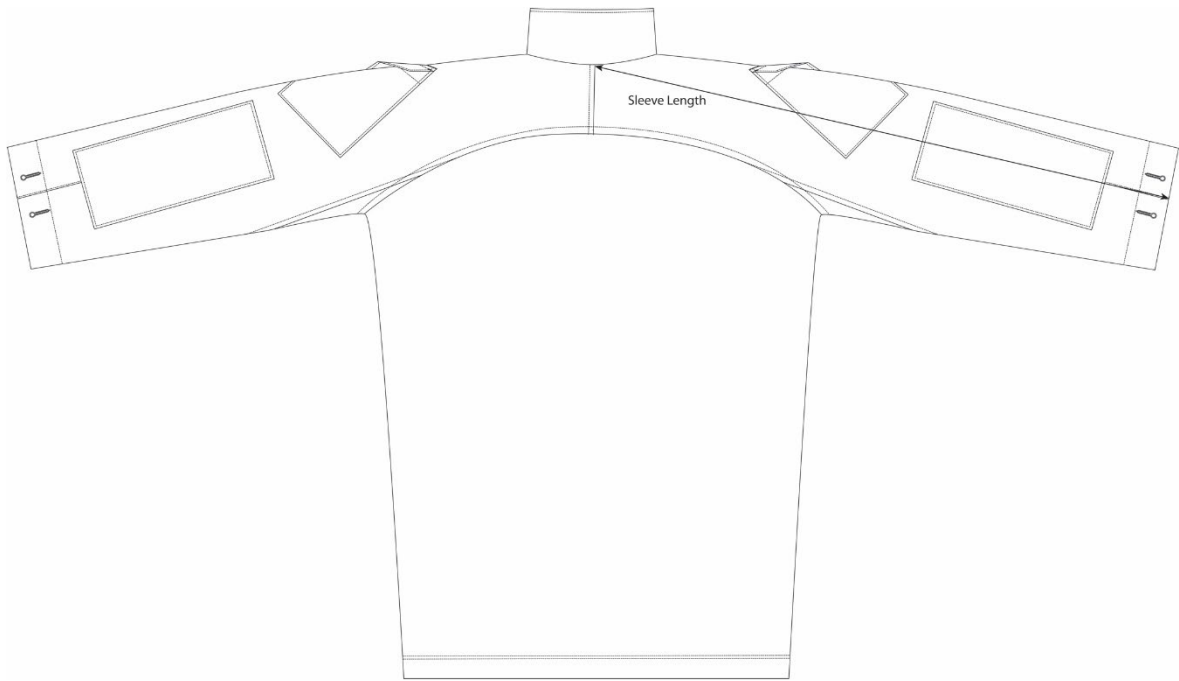


Figure 10. Sleeve Length Finished Dimension.

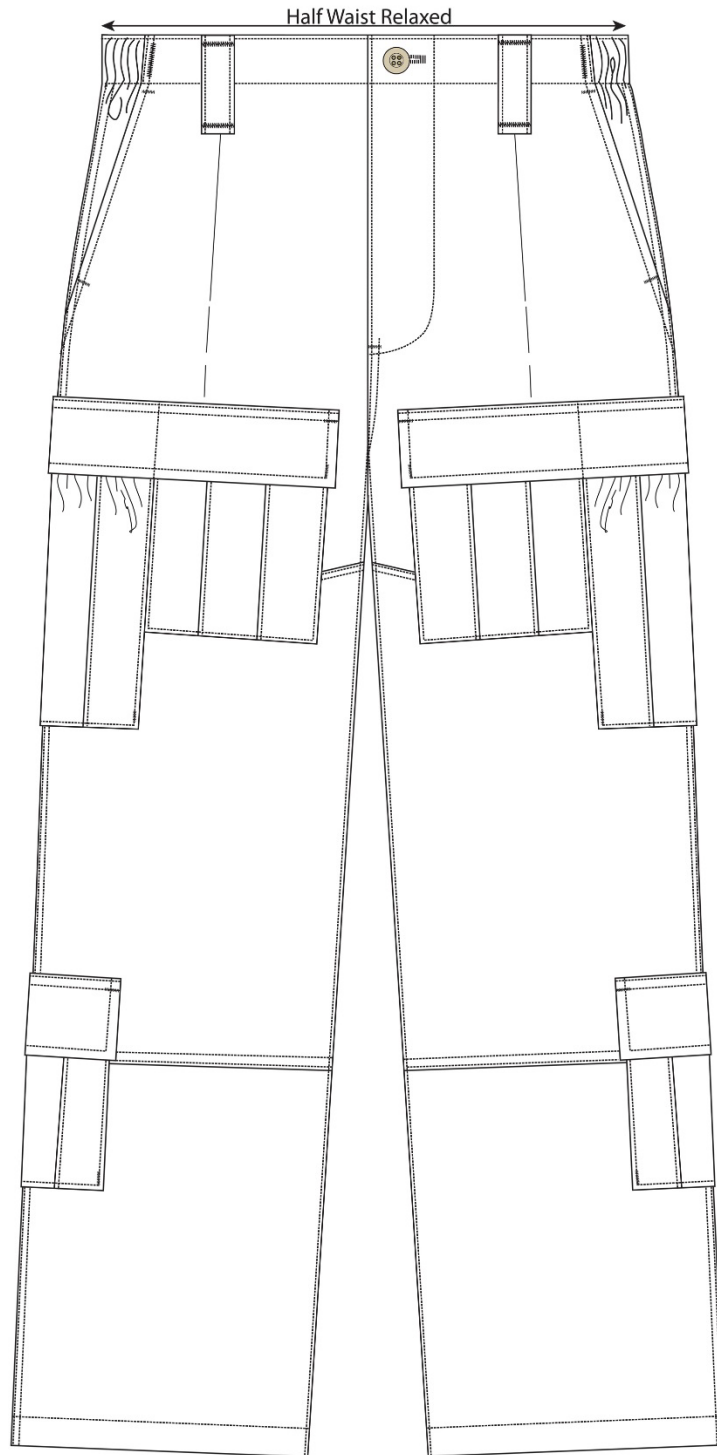


Figure 11. Half Waist Relaxed Finished Dimension.

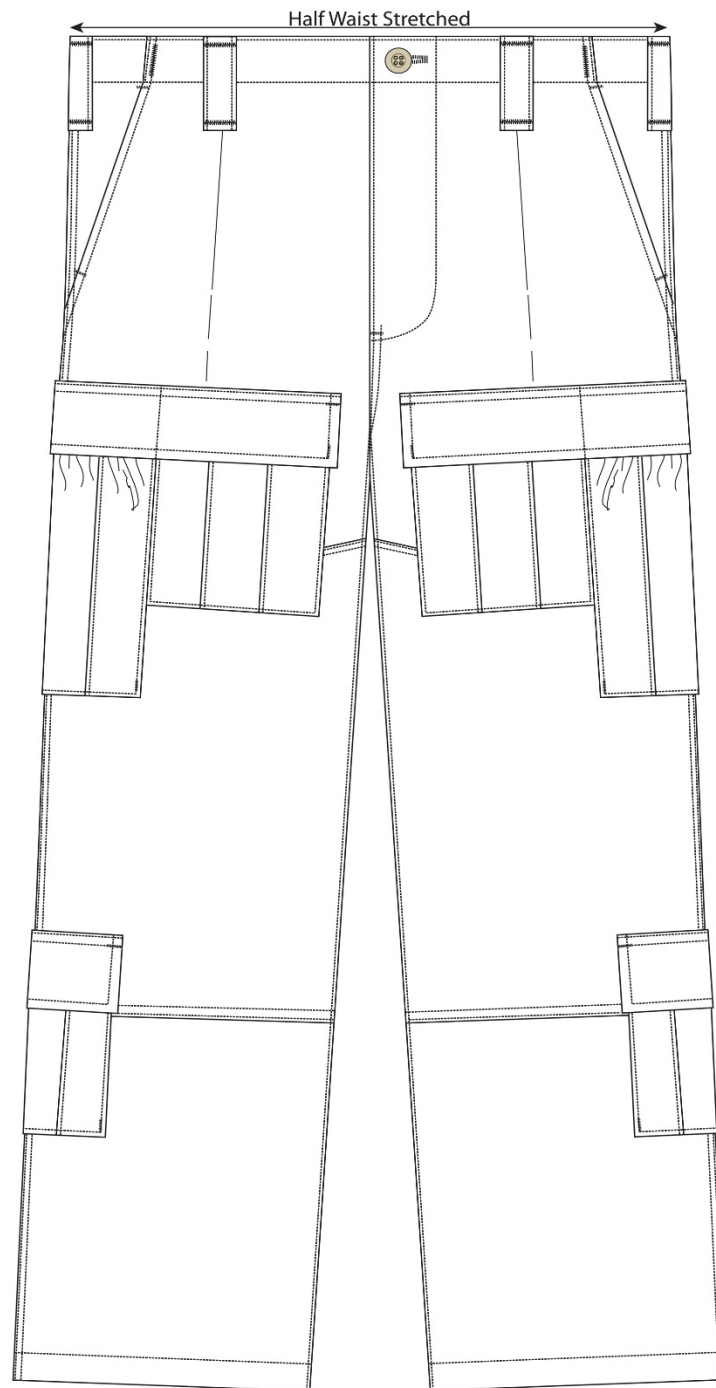


Figure 12. Half Waist Stretched Finished Dimension.



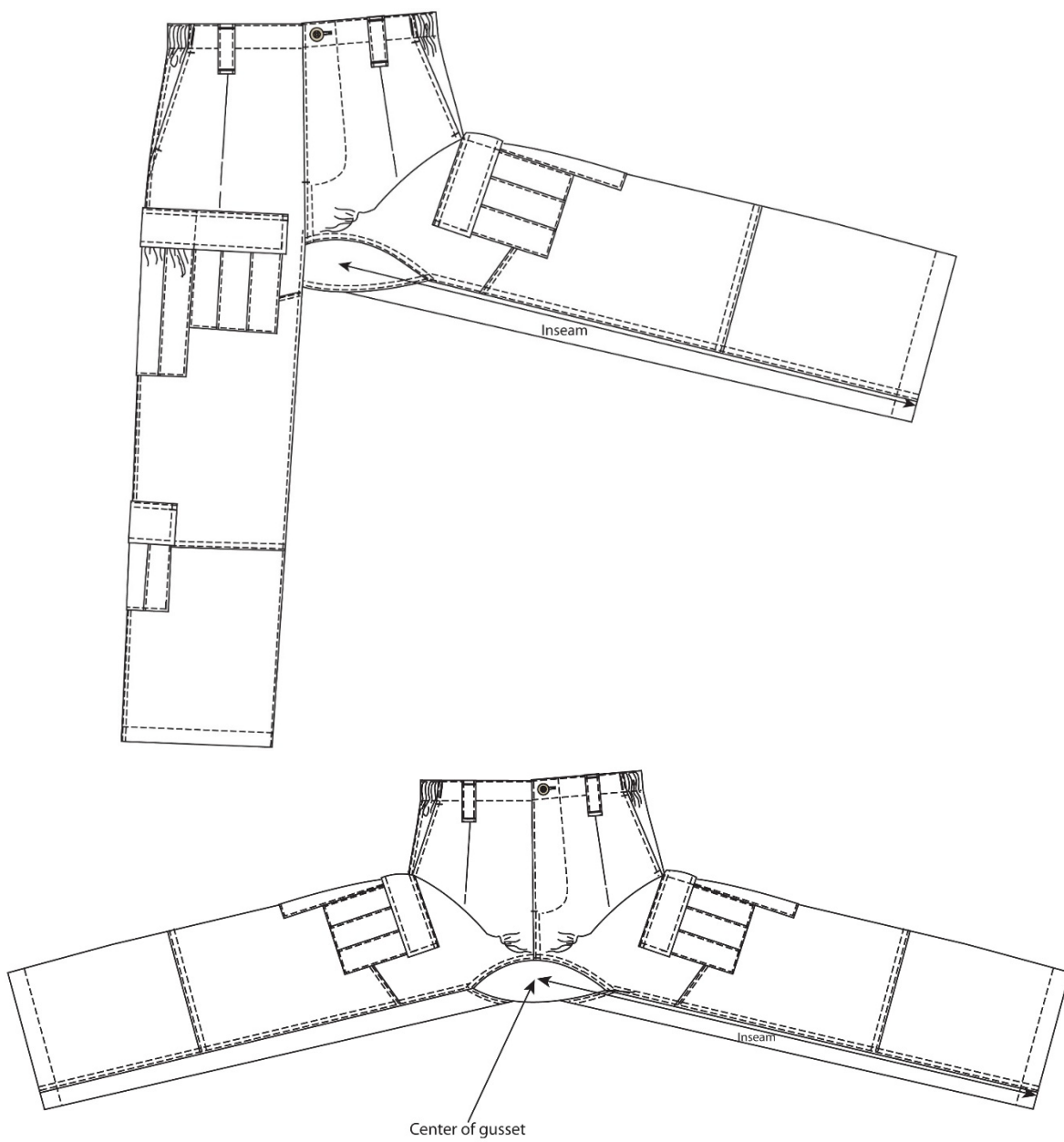


Figure 13. Trouser Inseam Finished Dimension.